

CC 101

Introduction to Computing — Complete Study Reference

Course: **CC 101**

Units: **3 units**

Year Level: **1st Year**

Type: **Pure Theory · No Coding**

CHAPTER 00

Course Overview & Study Priority Map

CC 101 is a **pure theory subject — zero programming involved**. That is CC 102's job. This course builds your mental model of computing: how hardware physically works, how data is encoded in binary, how software is classified, how networks connect the world, and your legal and ethical responsibilities as an IT professional in the Philippines. Every concept here becomes vocabulary you use for the rest of your 4-year BSIT program.

Critical Path — Layer 0 Gateway

CC 101 is a required prerequisite for **CC 103** (Intermediate Programming) and **PT 101** (Platform Technologies) in Semester 2. Those two subjects then unlock the majority of your 4-year BSIT program. Failing CC 101 delays everything downstream.

Exam Priority Map

Topic	Chapter	Exam Weight	Difficulty
Number Systems & Conversions	04	Very High	Medium

Topic	Chapter	Exam Weight	Difficulty
Computer Generations, History & Types		High	Easy
CPU Components & Hardware	03	High	Easy
OS Types & Functions	06	Medium	Easy
Data Representation	05	Medium	Easy
Software Classification	07	Medium	Easy
Networking Basics & OSI Model	08	Medium	Medium
IT Industries & Professions	01	Lower	Easy
Information System Types	09	Lower	Easy
Security & Philippine Laws	10	Lower	Easy
Graphic Design Principles & Canva	11	Practical/ Graded Output	Easy
Logic Gates & Boolean Algebra	12	High	Medium
Web Basics (Internet, WWW, HTML/XML)	13	Medium	Easy

CHAPTER 01

IT Industries & Profession

Before diving into hardware and theory, CC 101 opens with the bigger picture: what does a career in IT actually look like? This chapter distinguishes a job from a profession, maps the major categories of IT careers, and surveys where computers show up across every industry — the same framing QCU uses in Week 2 of the course.

| Job vs. Profession

A recurring exam-style question in this topic is simply: **what is the difference between a job and a profession?**

NARROWER CONCEPT

Job

A much narrower concept than a profession. A profession creates the place for a job — in other words, the profession comes first, and jobs exist within it.

BROADER CONCEPT

Profession

Indicates a broad field. A profession is something we have studied for — it requires educational qualifications as well as training, unlike a job, which doesn't necessarily require either.

| The Four Categories of IT Careers

Every IT career opportunity falls under one of four broad categories, grouped by the main focus of the job.

CATEGORY 1

Sales & Support Careers

Technical servicing: installation and maintenance of hardware and software.

Technical support: responding to phone and e-mail requests for customer assistance.

Technical sales: educating potential customers about a company's products so they want to buy them.

Technical training: teaching individuals how to use systems.

Technical writing: turning engineers'

CATEGORY 2

Software Development Careers

Software development: creation of new software products for commercial sale.

Systems programming: highly specialized work writing instructions in C++, JavaScript, or other languages to accomplish a succession of tasks.

Multimedia development: using computers to create images, sounds, and movies.

Quality assurance (QA): testing new products before they're released to the public.

technical skills and knowledge into written documentation.

CATEGORY 3

Application Development Careers

Application development: uses commercial software to build a system for a specific organization.

Systems analysis: gathering requirements and developing a design for systems.

Database administration: particularly important since the database sits at the core of a company's information systems.

Network administration: operations and maintenance of a company network.

Web administration: keeping a website working and up-to-date.

CATEGORY 4

Hardware Design Careers

Hardware design: the specification of computer hardware such as CPUs, video cards, and memory chips. Many hardware designers are electrical engineers, and many are computer engineers.

Traditional & Emerging IT Professions

Beyond the four broad categories, QCU's lecture calls out specific traditional and newly emerging IT roles you'll encounter in the field.

Network Engineer

TRADITIONAL

Keeps the network running

Technical expert responsible for setting up computer networks, maintaining them, and offering technical support to users where necessary.

Software Engineer

TRADITIONAL

Builds the software life cycle

Develops the fundamental concepts that exist within the software life cycle, from design through delivery.

Quality Assurance Engineer

TRADITIONAL

Guards the standard

Monitors every phase of the development process to ensure the design and software adhere to company standards.

IT Project Manager

Electronic Engineer

UI/UX Designer

TRADITIONAL

Leads projects end-to-end

Responsible for leading a project from its inception all the way to execution.

EMERGING

Designs everyday devices

Creates, designs, and develops everyday devices such as mobile phones, portable music devices, and computers.

EMERGING

UI = navigation, UX = experience

UI designers are primarily concerned with how a user navigates through a digital product. **UX designers** are responsible for creating an optimal experience for the user when interacting with a digital or physical product — even something like a coffee machine.

Business Analyst

EMERGING

Bridges IT and the business

Uses data analytics to assess processes, determine requirements, and deliver data-driven recommendations and reports to executives and stakeholders.

▪

Software/QA Architect

EMERGING

Senior QA leadership

A senior position responsible for the Software Quality Assurance team — leading QA initiatives and managing QA processes, test methods, risk analysis, and resource planning for client projects.

Fastest-Growing IT Careers

Among the top 15 fastest-growing positions overall: **computer software developers, computer support specialists, network and computer systems administrators, network systems and data communications analysts, and database administrators.**

Uses of Computers Across Industries

Computers are no longer confined to IT departments — they're embedded into nearly every field and industry.

Field / Industry	How Computers Are Used
Education	Computer-aided instruction, digital classrooms, curriculum development tools.
Health & Medicine	Diagnostic imaging (CT/MRI scanners), patient records, monitoring equipment in hospitals.
Financial Institutions	Online banking, account management, credit card and payment processing.
Pass Time / Entertainment	Video games, streaming, esports — computers as a leisure and entertainment medium.
Transport System	Onboard train/vehicle computers, traffic control, railway certification systems (e.g., EN 50155-certified railway computers).
Business & Corporate	Reporting, presentations, collaboration, and decision-making — computers are now inevitable at every corporate stage.
E-Commerce	Online ordering, food delivery apps, digital storefronts accessible from any device.
Defense	Command-and-control systems, satellite tracking, simulation and monitoring stations.
Design	Digital illustration, fashion design, and creative production software replacing traditional manual tools.

CHAPTER 02

History, Generations & Types of Computers

Computers evolved over thousands of years from simple counting tools to quantum processors. This chapter traces that full evolution — the inventors, machines, and technological leaps that define each generation — then

covers the different types of computers in use today. Expect exam questions on key names, dates, what defined each generation, and how each computer type differs.

Pre-Computer Era: Ancient Tools

~3000 BC

Abacus — Mesopotamia & China

Earliest known computing device. Beads on rods represent number columns. Used for arithmetic. Still used in some Asian schools for mental math training. Remarkably fast in trained hands.

~150 BC

Antikythera Mechanism — Ancient Greece

Recovered from a Greek shipwreck. A bronze-gear analog device used to predict astronomical positions and eclipses. Often called the world's first analog computer. Its complexity wasn't matched for over 1,400 years.

1450

Movable-Type Printing Press — Johann Gutenberg, Germany

Marks the start of the "Mechanical Age of Computing" (1450–1840) in QCU's official lecture sequence. Gutenberg's movable metal-type printing process triggered the first information explosion in history, making the mass production and spread of written knowledge possible for the first time.

1617

Napier's Bones — John Napier, Scotland

Numbered rods used to simplify multiplication and division. Napier also invented logarithms, which revolutionized scientific calculation. The rods were engraved with multiplication tables cut into sections to allow rapid multiplication.

1642

Pascaline — Blaise Pascal, France

Age 19 when he built it

First true mechanical calculator, built to help his father with tax computations. Could add and subtract using interlocking gears. About 50 were built. Pascal is also the namesake of the Pascal programming language.

1673

Stepped Reckoner — Gottfried Leibniz, Germany

Improved on Pascal's machine to also perform multiplication and division. Leibniz also advocated for the binary number system — the same system all modern computers use today.

1804

Jacquard's Loom — Joseph Marie Jacquard, France

Simplified the manufacturing of textiles with complex patterns (brocade, damask, matelassé) using punched cards with punched holes — each row of holes corresponding to one row of the fabric design. Its punch-card control system directly inspired the input method Babbage later used for the Analytical Engine.

1822

Difference Engine — Charles Babbage, England

Designed (never fully completed during his lifetime) to compute mathematical tables automatically, eliminating human calculation errors. This machine is what earns Babbage the title "Father of the Computer."

1837

Analytical Engine — Charles Babbage & Ada Lovelace

Babbage's more ambitious design: a general-purpose mechanical computer with separate memory ("store") and processor ("mill"), capable of conditional branching and loops. **Ada Lovelace** wrote notes on this machine including what is recognized as the first algorithm intended for a machine — making her the world's first programmer.

1854

Boolean Algebra — George Boole, England

Boole published "An Investigation of the Laws of Thought," establishing Boolean algebra: a system of logic using only TRUE and FALSE (1 and 0). This mathematical foundation is what makes all digital computing possible — every computer operation reduces to Boolean logic.

1890

Punch Card Tabulator — Herman Hollerith, USA

Built for the 1890 US Census. Used electrical sensing to tabulate data on punched cards — cutting census processing from an estimated 8 years to just 18 months. Hollerith founded the Tabulating Machine Company, which eventually became IBM.

1936

Turing Machine — Alan Turing, England

Not a physical machine but a theoretical model: a device that reads/writes symbols on an infinite tape according to a set of rules. Turing proved this simple machine could compute anything computable — laying the theoretical foundation of all computer science. Turing also broke the Nazi Enigma code in WWII, shortening the war.

1945

Von Neumann Architecture — John von Neumann, USA

Published the "stored program" concept: a computer's instructions AND data should both be stored in the same memory. This architecture (CPU + Memory + I/O + stored programs) is still the blueprint of virtually every computer built today, 80 years later.

1945

ENIAC — Eckert & Mauchly, USA

Electronic Numerical Integrator and Computer — first general-purpose electronic digital computer. Facts: 17,468 vacuum tubes, 70,000 resistors, 30 tons, filled an 1,800 sq ft room, consumed 150 kilowatts. Used to calculate artillery firing tables and hydrogen bomb calculations.

1951

UNIVAC I — Remington Rand, USA

First commercial computer sold in the United States. Used for the 1950 US Census. Made famous when CBS used it to correctly predict Eisenhower's 1952 presidential election victory — the first computer prediction broadcast on television.

The Five Generations of Computers

Computer generations are defined by the key **electronic component** used. Each generation brought dramatic improvements in size, speed, cost, and capability.

1965 — 1970

3rd Generation — Integrated Circuits

Technology: ICs — multiple transistors on one silicon chip (Jack Kilby, Texas Instruments, 1958)

Language: High-level languages dominant: BASIC, COBOL, PL/I

I/O: Keyboards and monitors introduced (goodbye punch cards)

1971 — Present

4th Generation —

Technology: Very Large Scale Integration (VLSI)

First: Intel 4004 (1971)

Language: All high-level languages

Key: Apple II (1977), IBM PC (1981)

Storage: Magnetic disks (HDDs)
Size: Minicomputers introduced — desk-sized
Speed: Nanoseconds
Key Machines: IBM System/360, PDP-8 (first minicomputer)

Speed: Picoseconds
Impact: Personal computers











Memory Trick: 5 Generations

V-T-I-M-A — Vacuum tubes → Transistors → Integrated Circuits → Microprocessors → Artificial Intelligence. Each solved the previous generation's biggest problem: heat, size, cost, access, and intelligence.

Types of Computers

Computers are also classified by **size, processing power, and intended use** rather than just by generation. This classification is a direct QCU learning outcome for Week 3 — expect exam items asking you to match a description to its correct type.

Type	Description
Microcomputer	A personal computer designed to meet the computing needs of a single individual. Provides access to a wide range of applications: word processing, photo editing, e-mail, and internet browsing.
Desktop Microcomputer	A microcomputer that fits on a desk and runs on power from an electrical wall outlet. Its CPU can be housed in either a vertical or horizontal case, with separate components (keyboard, mouse, etc.) plugged into the unit.
Laptop Computer	A portable, compact computer that runs on either an electrical wall outlet or a battery unit. All components are built into one compact unit. Usually more expensive than a comparable desktop. Also called a Notebook.
Workstation	A powerful desktop computer designed for specialized, processing-heavy tasks. Can also be an ordinary personal computer attached to a LAN (local area network).
Supercomputer	The fastest computer in the world at the time it was built. Tackles tasks impractical for

Type	Description
	other computers, such as breaking codes and modeling weather systems.
Mainframe	A large, expensive computer capable of simultaneously processing data for hundreds or thousands of users. Used to store, manage, and process large amounts of data that must be reliable, secure, and centralized. Usually housed in a closet-sized cabinet.
Server	A computer whose purpose is to "serve" — supplying data to its users, usually through a LAN (local area network).
Handheld (PDA)	Short for Personal Digital Assistant. A computer that fits into a pocket, runs on batteries, and is held in the hand while in use. Typically functions as an appointment book, address book, calculator, and notepad, and can be synced with a personal microcomputer as a backup.

Key People in Computing History

Charles Babbage

1791–1871 · England

Father of the Computer

Designed the Difference Engine and Analytical Engine — the first designs for a general-purpose mechanical computer. Never fully built in his lifetime but conceptually correct.

Ada Lovelace

1815–1852 · England

World's First Programmer

Wrote the first algorithm intended to be processed by a machine (for Babbage's Analytical Engine). Recognized that machines could go beyond pure calculation. The Ada language is named after her.

Alan Turing

1912–1954 · England

Father of Computer Science

Created the Turing Machine theoretical model. Broke the Nazi Enigma code in WWII, arguably shortening the war by 2 years. Proposed the Turing Test for AI. The ACM Turing Award is the "Nobel Prize of Computing."

John von Neumann

1903–1957 · Hungary/USA

Architect of Modern Computing

Grace Hopper

1906–1992 · USA

Mother of COBOL

Linus Torvalds

1969–present · Finland

Creator of Linux

Proposed the stored-program concept — instructions and data in the same memory. The von Neumann architecture is still the blueprint of virtually all computers built today.

US Navy Rear Admiral. Created the first compiler (A-0, 1952) and co-designed COBOL — still running billions of dollars of banking transactions today. Popularized the term "debugging" after finding an actual moth in a relay.

Created the Linux kernel in 1991 as a 21-year-old student. Linux now powers Android phones, 96% of the world's servers, supercomputers, and internet infrastructure. Your Debian setup IS Linux.

Tim Berners-Lee

1955–present · England

Inventor of the World Wide Web

Invented the WWW in 1989 at CERN — not the Internet itself (that already existed). Created HTTP, HTML, and the URL system. Made it royalty-free so it became universal.

Bill Gates

1955–present · USA

Co-founder of Microsoft

Co-founded Microsoft with Paul Allen in 1975. MS-DOS and Windows made personal computing mainstream. Microsoft Office became the global standard for office productivity.

Steve Jobs

1955–2011 · USA

Co-founder of Apple

Co-founded Apple with Steve Wozniak in 1976. Revolutionized personal computers (Mac), music (iPod), phones (iPhone), and tablets (iPad). Known for his uncompromising focus on design and user experience.

CHAPTER 03

Computer Hardware

Hardware refers to all **physical components** you can touch. Every computer — from ENIAC to a modern smartphone — follows the same fundamental architecture: a processor, memory, storage, and input/output devices connected by buses. This chapter breaks each component down in detail.

The Computer System = Hardware + Software + Peopleware

A "computer system" is more than the machine itself — it's a complete, working setup. It has three parts: **Hardware** (physical, touchable components — this whole chapter), **Software** (the instructions that tell hardware what to do — see Chapter 07), and **Peopleware** (the human role — the people who operate, program, and maintain the system, covered at the end of this chapter). Remove any one part and the "system" stops functioning.

Von Neumann Architecture — The Blueprint of All Computers

All modern computers follow this structure: **CPU** ↔ connected via a **Bus** ↔ **Memory (RAM)** ↔ **Storage (SSD/HDD)** ↔ **I/O Devices**. Instructions AND data both live in the same memory. The CPU fetches instructions, decodes them, and executes them in a continuous loop called the Fetch-Decode-Execute cycle.

CPU — Central Processing Unit

The CPU is the **"brain" of the computer**. It performs all logical and arithmetic operations and controls all other hardware. Everything the computer does passes through the CPU.

Component	Full Name	What It Does
CU	Control Unit	Directs all CPU operations. Fetches instructions from memory, decodes them, and tells the ALU and registers what to do. The "manager" or "conductor" of the CPU — it does no actual calculation itself.
ALU	Arithmetic Logic Unit	Performs ALL arithmetic (addition, subtraction, multiplication, division) and ALL logical operations (AND, OR, NOT, XOR, comparisons like >, <, ==). The "calculator" of the CPU.
Registers	Processor Registers	Ultra-fast, tiny storage locations INSIDE the CPU

Component	Full Name	What It Does
		core itself. Hold data currently being processed. Much faster than RAM. Examples: Program Counter (PC) tracks next instruction address; Accumulator stores results; Instruction Register holds current instruction.
Cache	CPU Cache Memory	Small but very fast memory between CPU and RAM. Stores frequently used data to avoid slow RAM access. L1 (fastest, smallest, inside each core ~32-512KB), L2 (medium), L3 (largest, shared between cores, 4-64MB).
Clock	System Clock	Generates synchronized pulses that time all CPU operations. Measured in GHz (gigahertz = billions of pulses/second). Modern CPUs: 3-5.5 GHz. More GHz generally = faster processing, but architecture matters more now.
Cores	Processing Cores	Each core is an independent processing unit. Dual-core = 2 CPUs in one chip. Modern phones have 8 cores (octa-core). A Dimensity 8350-class chipset, for example, has 8 cores across 3 performance clusters (1+3+4 big.MIDDLE.LITTLE architecture).

Fetch-Decode-Execute Cycle

This is the fundamental loop the CPU runs billions of times per second:

1 FETCH

Control Unit reads the next instruction from RAM. The Program Counter register holds the memory address of the next instruction. After fetching, the PC increments to point to the next instruction.

2 DECODE

Control Unit decodes the instruction — translates machine code into control signals that tell the ALU and registers exactly what to do (add? compare? move data?).

3 EXECUTE

The ALU or other unit performs the operation. Result is stored back in a register or written to memory. Then the cycle repeats — billions of times per second.

Filipino Analogy — The CPU

Think of the CPU as the **QCU Student Affairs Office**. The Control Unit is the Director who reads requests (instructions) and tells staff what to do. The ALU is the Accounting staff doing all calculations. Registers = the director's desk (tiny, fast, only what's immediately needed). Cache = the office filing cabinet (nearby, fast). RAM = the main storage room. Storage = the Records archive in the basement.

Memory

RAM — Random Access Memory

RAM is the computer's **temporary working memory** — storage for data and programs currently in use. When you open Chrome, VS Code, and a music app at the same time, all three are loaded into RAM.

Property	Detail
Volatile?	YES — all data lost when power is cut. This is why unsaved work disappears in a crash.
Speed	Slower than CPU cache but much faster than SSD/HDD storage.
Types	DRAM (Dynamic RAM — uses capacitors, needs refresh), SRAM (Static RAM — faster, used in cache), DDR4/DDR5 (modern desktop), LPDDR4X/LPDDR5 (mobile — LP = Low Power). Most current mid-range and flagship phones use LPDDR5.

Property	Detail
Capacity effect	More RAM = more apps running simultaneously without slowdown or swapping to storage.

ROM — Read-Only Memory

ROM stores **permanent firmware** — non-volatile (survives power off). Contains the instructions the computer uses to start up before the OS loads.

Type	Full Name	Description
ROM	Read-Only Memory	Factory-programmed. Cannot be changed. Contains boot firmware.
PROM	Programmable ROM	Written once by manufacturer or user using a PROM writer tool.
EPROM	Erasable PROM	Erased with UV light and reprogrammed. Used in older microcontrollers.
EEPROM	Electrically Erasable PROM	Erased/reprogrammed electrically without removing from circuit. Used in BIOS chips.
Flash ROM	Flash Memory	Modern evolution of EEPROM. Used in SSDs, USB drives, phone storage, SD cards. Erased in "blocks."

Storage Devices

Storage is **non-volatile** — data persists after power off. This is where your OS, programs, photos, and files permanently live.

Type	Technology	Speed	Capacity	Durability	Use Case
HDD	Magnetic spinning			Fragile (moving	Bulk storage,

Type	Technology	Speed	Capacity	Durability	Use Case
	platters + moving read/write head	Slow (100–200 MB/s)	500 GB – 20 TB	parts, vulnerable to shock)	backups, cheap capacity
SSD (SATA)	NAND Flash — no moving parts	Fast (400–600 MB/s)	120 GB – 8 TB	Very durable	OS drives, programs
NVMe SSD	NAND Flash + PCIe interface	Very fast (3,000–7,000+ MB/s)	256 GB – 4 TB	Very durable	High-performance systems, gaming
USB Flash Drive	NAND Flash	Moderate (25–450 MB/s USB 3.x)	4 GB – 2 TB	Durable	Portable file transfer
SD Card	NAND Flash	Varies by class	2 GB – 2 TB	Durable	Cameras, phones
Optical Disc	Laser reading pits/lands on polycarbonate disc	Very slow	CD: 700MB · DVD: 4.7–8.5GB · Blu-ray: 25–100GB	Scratches easily	Legacy media, distribution

Input Devices

Input devices **send data INTO the computer** from the user or the environment.

Device	Type of Input	How It Works
Keyboard	Text / Commands	Key presses generate electrical signals mapped to character codes (ASCII/Unicode). Can be mechanical (key switches), membrane, or capacitive.

Device	Type of Input	How It Works
Mouse / Touchpad	Positional	Optical sensor tracks movement on a surface; generates relative X/Y coordinates. Buttons generate click signals.
Touchscreen	Touch / Positional	Capacitive sensors detect electrical change from finger contact. Converts touch coordinates to input. Modern smartphone touchscreens typically support up to 10 simultaneous touch points.
Microphone	Audio	Converts sound waves (analog) to digital audio signal via ADC (Analog-to-Digital Converter). Condenser, dynamic, or MEMS (in phones) types.
Webcam / Camera	Visual	CMOS image sensor captures light from a scene and converts it to digital pixel data.
Scanner	Document / Image	Optical sensor reads reflected light from a document, line by line, converting it to a digital image.
Fingerprint Reader	Biometric	Optical or capacitive sensor captures fingerprint pattern for authentication. In-display optical sensors are common in modern phones.
Barcode / QR Scanner	Data	Laser or camera reads bar/pattern, decodes product or data. Used at Jollibee, SM, and GCash QR payments.

Output Devices

Output devices **present data FROM the computer** to the user.

Device	Output Type	Key Details
LCD Monitor	Visual	Liquid Crystal Display — requires a backlight (LED). Pixels are formed by liquid crystals blocking or allowing light to pass through color filters. Most common desktop display type.
OLED / AMOLED Display	Visual	Each pixel generates its own light — no backlight needed. True blacks (pixel simply off). Better contrast ratio, deeper colors, thinner panels. A typical modern flagship phone runs an AMOLED display around 2720×1224 resolution at up to 144Hz.
Inkjet Printer	Physical print	Sprays microscopic ink droplets onto paper via print head nozzles. Good for photos. Ink is expensive per page. Nozzles can clog if unused.
Laser Printer	Physical print	Uses toner powder + heat (fuser unit) to bond toner to paper. Fast, cheap per page for text and documents. Common in offices and schools.
Speakers / Headphones	Audio	Convert digital audio (DAC → analog electrical signal) to physical sound waves via an electromagnet + vibrating cone/diaphragm.
Projector	Visual	DLP (Digital Light Processing) or LCD technology projects a

Device	Output Type	Key Details
		magnified image onto a surface. Common in QCU classrooms.

Common Ports & Connectors

Port	Max Speed	Primary Use
USB-A 3.0 (Blue port)	5 Gbps	Flash drives, keyboards, mice, chargers. The rectangular blue USB port on most laptops.
USB-C 3.2	20 Gbps	Modern universal port: high-speed data, video output, power delivery. Your phone uses USB-C.
HDMI 2.1	48 Gbps	Audio + video combined to monitors, TVs, projectors. Most common display connection.
DisplayPort 2.1	80 Gbps	High refresh rate monitor connection, preferred for PC gaming setups.
RJ-45 (Ethernet)	Up to 10 Gbps	Wired network connection. Globe/PLDT modem to laptop for reliable internet.
3.5mm Audio Jack	Analog	Headphones, microphones, speakers. TRS (3-ring = stereo) or TRRS (4-ring = stereo + mic).

The Motherboard — Where Everything Connects

The **motherboard** holds together every component covered above — CPU, memory, storage, and I/O — through a set of standardized sockets, slots, and connectors. Understanding these ties the whole chapter together: the CPU

sits in the socket, RAM sits in the memory slots, storage plugs into IDE/SATA, and expansion cards plug into PCI/AGP.

Component	What It Does
Socket	The connection point that allows a CPU to be physically attached to the motherboard (e.g. older Socket 370/PGA370, modern LGA sockets). The socket type must match the CPU generation.
ATX Power Connector	The connector(s) inside the case that feed power from the PSU (power supply unit) to an ATX-style motherboard — commonly a 24-pin main connector plus a separate 4-pin CPU power connector.
Memory Slot (RAM Slot)	Lets RAM modules be inserted into the motherboard. Most boards have 2-4 slots; the slot type determines which RAM generation (DDR3/DDR4/DDR5) is compatible.
DIMM	Dual In-line Memory Module — the circuit board that physically holds the memory chips and plugs into a memory slot. A 64-bit data path lets DIMMs be installed one at a time, unlike older SIMMs which had to be added in pairs.
IDE / ATA	Integrated Drive Electronics (also called Advanced Technology Attachment) — an older standard interface connecting the motherboard to storage drives (HDDs, CD/DVD drives) via a ribbon cable, superseded on modern boards by SATA.
Northbridge	An integrated circuit that handles high-speed communication between the CPU, RAM, and the graphics interface (AGP/PCIe). Directly connected to the CPU — the "closer, faster" chipset.
Southbridge	An integrated circuit responsible for slower I/O: hard drive controller, USB, PCI, legacy IDE, BIOS, and Ethernet. Communicates with the CPU through the Northbridge.
PCI Slot	

Component	What It Does
	Peripheral Component Interconnect — a general-purpose expansion slot for adding cards like graphics cards, modems, or extra USB ports.
AGP Slot	Accelerated Graphics Port — an older dedicated point-to-point slot built specifically for video cards and 3D accelerators, giving the graphics controller direct access to system memory. Mostly replaced by PCI Express (PCIe) on modern boards.

Filipino Analogy — The Motherboard

If the CPU is the Student Affairs Director, the **motherboard is the entire QCU San Bartolome campus building** — it doesn't do the work itself, but every office (component) has a designated room (slot/socket/port) and a hallway (bus) connecting it to every other office. Northbridge is the fast express hallway connecting the Director's office directly to Records (RAM) and the Design Office (graphics). Southbridge is the general corridor connecting everything else — printers, gate scanners, and the back-office archive (storage).

Peopleware & Types of Computer Users

Peopleware refers to the human role in an IT system — the people who use, program, and maintain the hardware and software. It treats human talent as a key planning resource in any technical business model, just as important as the physical and logical components.

User Type	Description
Casual / Regular User	Uses a computer for everyday, general-purpose tasks — browsing, messaging, schoolwork, entertainment. Needs no specialized technical knowledge.
Small Office/Home Office (SOHO)	Uses computing for small-scale business or freelance needs — invoicing, light design work, basic networking between a few devices.

User Type	Description
Power User	Uses a computer for demanding, specialized, or technical tasks — programming, video editing, gaming, system administration — and typically understands hardware/software at a deeper level than a casual user.

CHAPTER 04

Number Systems & Conversions

This is the **most exam-heavy topic in CC 101**. Number systems are ways to represent numbers using different bases. Computers use binary internally, programmers use hexadecimal as shorthand, and humans use decimal. Master all conversions between the four systems.

Exam Alert

Conversion problems — especially decimal ↔ binary and binary ↔ hexadecimal — appear on virtually every CC 101 midterm and final. Do NOT just memorize the steps; understand the pattern so you can solve any number, not just the examples you studied.

The Four Number Systems at a Glance

System	Base	Digits Used	Example	Used In
Binary	2	0, 1	$1010_2 = 10_{10}$	All internal computer operations — transistors have exactly two states: ON or OFF
Octal	8	0 1 2 3 4 5 6 7	$12_8 = 10_{10}$	Older UNIX file permissions (chmod 755),

System	Base	Digits Used	Example	Used In
				shorthand for binary groups of 3
Decimal	10	0 1 2 3 4 5 6 7 8 9	10_{10}	Human everyday counting
Hexadecimal	16	0–9, A B C D E F	$A_{16} = 10_{10}$	Memory addresses, HTML/CSS color codes (#FF5733), machine code shorthand

Why Do Computers Use Binary?

A transistor is a switch — it is either ON (conducting electricity) or OFF (blocking it). These two states map perfectly to 1 and 0. Every computation reduces to billions of these ON/OFF decisions happening billions of times per second. Binary is the only number system a physical transistor can naturally represent.

Hexadecimal Digit Reference Chart — MEMORIZE THIS

Decimal	Binary (4-bit)	Hex	Decimal	Binary (4-bit)	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	A
3	0011	3	11	1011	B
4	0100	4	12	1100	C
5	0101	5	13	1101	D
6	0110	6	14	1110	E

Decimal	Binary (4-bit)	Hex	Decimal	Binary (4-bit)	Hex
7	0111	7	15	1111	F

Conversion Methods — Full Step-by-Step Examples

1. Binary → Decimal (Positional Value Method)

Each binary digit is multiplied by 2 raised to the power of its position (counting right-to-left starting at 0), then all products are summed.



EXAMPLE: $1011_2 \rightarrow \text{DECIMAL}$

Positions (right to left): 3 2 1 0

Digits: 1 0 1 1

$$1 \times 2^3 = 1 \times 8 = 8$$

$$0 \times 2^2 = 0 \times 4 = 0$$

$$1 \times 2^1 = 1 \times 2 = 2$$

$$1 \times 2^0 = 1 \times 1 = 1$$

$$\text{Sum: } 8 + 0 + 2 + 1 = 11$$

$$\therefore 1011_2 = 11_{10} \quad \checkmark$$



EXAMPLE 2: $11010110_2 \rightarrow \text{DECIMAL}$

Positions: 7 6 5 4 3 2 1 0

Digits: 1 1 0 1 0 1 1 0

$$1 \times 128 = 128$$

$$1 \times 64 = 64$$

$$0 \times 32 = 0$$

$$1 \times 16 = 16$$

$$0 \times 8 = 0$$

$$1 \times 4 = 4$$

$$1 \times 2 = 2$$

$$0 \times 1 = 0$$

$$\text{Sum: } 128 + 64 + 0 + 16 + 0 + 4 + 2 + 0 = 214$$

$$\therefore 11010110_2 = 214_{10} \quad \checkmark$$

2. Decimal $\rightarrow$ Binary (Repeated Division by 2)

Divide the number by 2 repeatedly, recording remainders. Read remainders **BOTTOM to TOP** for the binary result.



EXAMPLE: $25_{10} \rightarrow \text{BINARY}$

```
25 ÷ 2 = 12 remainder 1 ← LSB (Least Significant Bit, rightmost)
12 ÷ 2 = 6  remainder 0
6 ÷ 2 = 3   remainder 0
3 ÷ 2 = 1   remainder 1
1 ÷ 2 = 0   remainder 1 ← MSB (Most Significant Bit, leftmost)
```

Read remainders BOTTOM to TOP: 1, 1, 0, 0, 1

$$\therefore 25_{10} = 11001_2$$

$$\text{Verify: } 16 + 8 + 0 + 0 + 1 = 25 \quad \checkmark$$



EXAMPLE 2: $156_{10} \rightarrow \text{BINARY}$

```
156 ÷ 2 = 78 R 0
78 ÷ 2 = 39 R 0
39 ÷ 2 = 19 R 1
19 ÷ 2 = 9  R 1
9 ÷ 2 = 4   R 1
4 ÷ 2 = 2   R 0
2 ÷ 2 = 1   R 0
1 ÷ 2 = 0   R 1
```

Bottom to top: 10011100

$$\therefore 156_{10} = 10011100_2$$

$$\text{Verify: } 128 + 0 + 0 + 16 + 8 + 4 + 0 + 0 = 156 \quad \checkmark$$

Alternate Method: Powers of Two (Subtraction Approach)

Instead of dividing, you can build the binary number by repeatedly subtracting the largest power of 2 that still fits. Useful when you have the powers-of-2 sequence memorized: ...**256, 128, 64, 32, 16, 8, 4, 2, 1**.



EXAMPLE: $300_{10} \rightarrow \text{BINARY (POWERS OF TWO)}$

```
256 fits into 300 → write 1, remainder 300-256 = 44
128 does NOT fit into 44 → write 0
64 does NOT fit into 44 → write 0
32 fits into 44 → write 1, remainder 44-32 = 12
16 does NOT fit into 12 → write 0
8 fits into 12 → write 1, remainder 12-8 = 4
4 fits into 4 → write 1, remainder 4-4 = 0
2 does NOT fit into 0 → write 0
1 does NOT fit into 0 → write 0
```

```
256 128 64 32 16 8 4 2 1
1   0   0  1  0  1  1  0  0
```

$\therefore 300_{10} = 100101100_2$

Verify: $256+32+8+4 = 300 \checkmark$

Division vs. Powers of Two

Both methods always give the same answer — use whichever is faster for you. Division-by-2 is more mechanical (good for large or awkward numbers); Powers of Two is faster once you have the sequence 1, 2, 4, 8, 16, 32, 64, 128, 256... memorized.

3. Binary → Hexadecimal (Group into 4s)

Group binary digits into sets of **4 from the right**. Pad with leading zeros if needed. Convert each group to its hex digit.



EXAMPLE: $10110111_2 \rightarrow \text{HEX}$

```
Group into 4s from right:
1011 | 0111
```

Convert each group using the chart:

$$1011 = 8+2+1 = 11 = B$$

$$0111 = 4+2+1 = 7 = 7$$

$$\therefore 10110111_2 = B7_{16}$$



EXAMPLE 2: $1111001011_2 \rightarrow \text{HEX (PAD LEFT WITH 0S)}$

Pad to groups of 4: 0011 | 1100 | 1011

Convert:

$$0011 = 2+1 = 3$$

$$1100 = 8+4 = 12 = C$$

$$1011 = 8+2+1 = 11 = B$$

$$\therefore 1111001011_2 = 3CB_{16}$$

4. Hexadecimal $\rightarrow$ Binary (Each hex digit = 4 binary digits)



EXAMPLE: $4F3_{16} \rightarrow \text{BINARY}$

$$4 \rightarrow 0100$$

$$F \rightarrow 1111 \quad (F = 15)$$

$$3 \rightarrow 0011$$

$$\therefore 4F3_{16} = 010011110011_2$$

5. Decimal $\rightarrow$ Hexadecimal (Repeated Division by 16)

Same as decimal-to-binary but divide by 16. If remainder ≥ 10 , convert to A-F.



EXAMPLE: $255_{10} \rightarrow \text{HEX}$

$$255 \div 16 = 15 \quad \text{remainder } 15 = F$$

$$15 \div 16 = 0 \quad \text{remainder } 15 = F$$

Read bottom to top: F, F

$$\therefore 255_{10} = FF_{16}$$

This is why pure white in web colors is #FFFFFF
(R=255, G=255, B=255 → FF FF FF)



EXAMPLE 2: $3072_{10} \rightarrow \text{HEX}$

```
3072 ÷ 16 = 192 R 0
192 ÷ 16 = 12 R 0
12 ÷ 16 = 0 R 12 = C
```

Read bottom to top: C, 0, 0

$\therefore 3072_{10} = C00_{16}$

6. Binary → Octal (Group into 3s)



EXAMPLE: $101011_2 \rightarrow \text{OCTAL}$

Group into 3s from right:

101 | 011

Convert:

101 = $4+1 = 5$

011 = $2+1 = 3$

$\therefore 101011_2 = 53_8$

Verify: $5 \times 8 + 3 = 43_{10}$

Binary: $32+8+2+1 = 43_{10} \checkmark$

7. Octal → Binary (Each Octal Digit = 3 Binary Digits)



EXAMPLE: $257_8 \rightarrow \text{BINARY}$

2 → 010

5 → 101

7 → 111

$\therefore 257_8 = 010101111_2$

8. Decimal → Octal (Repeated Division by 8)



EXAMPLE: $128_{10} \rightarrow \text{OCTAL}$

$$128 \div 8 = 16 \quad \text{remainder } 0$$

$$16 \div 8 = 2 \quad \text{remainder } 0$$

$$2 \div 8 = 0 \quad \text{remainder } 2$$

Read bottom to top: 2, 0, 0

$$\therefore 128_{10} = 200_8$$



EXAMPLE 2: $287_{10} \rightarrow \text{OCTAL}$

$$287 \div 8 = 35 \quad \text{remainder } 7$$

$$35 \div 8 = 4 \quad \text{remainder } 3$$

$$4 \div 8 = 0 \quad \text{remainder } 4$$

Read bottom to top: 4, 3, 7

$$\therefore 287_{10} = 437_8$$

9. Octal → Decimal (Positional Value Method)



EXAMPLE: $176_8 \rightarrow \text{DECIMAL}$

$$6 \times 8^0 = 6 \times 1 = 6$$

$$7 \times 8^1 = 7 \times 8 = 56$$

$$1 \times 8^2 = 1 \times 64 = 64$$

$$\text{Sum: } 6 + 56 + 64 = 126$$

$$\therefore 176_8 = 126_{10}$$

10. Octal → Hexadecimal (Convert via Binary)

There's no direct shortcut — convert octal to binary first, then regroup the binary into 4-bit chunks for hex.



EXAMPLE: $176_8 \rightarrow \text{HEXADECIMAL}$

Step 1: Each octal digit → 3-bit binary

1 → 001 7 → 111 6 → 110

Binary: 001111110

Step 2: Regroup into 4-bit chunks from the right

0001 1111 0 → pad → 0000 1111 10 → correctly grouped: 0111 1110

(equivalently: $176_8 = 126_{10} = 7E_{16}$, verified against Step 9 above)

∴ $176_8 = 7E_{16}$

11. Hexadecimal → Octal (Convert via Binary)



EXAMPLE: $176_{16} \rightarrow \text{OCTAL}$

Step 1: Each hex digit → 4-bit binary

1 → 0001 7 → 0111 6 → 0110

Binary: 000101110110

Step 2: Regroup into 3-bit chunks from the right

000 101 110 110 → drop leading zeros → 101 110 110

101 = 5 110 = 6 110 = 6

∴ $176_{16} = 566_8$

Key Insight — Octal/Hex Never Convert Directly

Octal (base 8 = 2^3) and Hexadecimal (base 16 = 2^4) both "line up" cleanly with binary because 8 and 16 are powers of 2 — but not with each other.

Always use **binary as the bridge** when converting between octal and hexadecimal.

Fractional (Radix Point) Conversions

Every number system has a "point" separating the whole part from the fractional part — decimal point, binary point, hexadecimal point. Digits to the **right** of the point use **negative exponents** (2^{-1} , 2^{-2} , 2^{-3} ...).

Binary → Decimal (With Fraction)



EXAMPLE: $1011.11_2 \rightarrow \text{DECIMAL}$

$$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$= 8 + 0 + 2 + 1 + 0.5 + 0.25$$

$$\text{Sum: } 8 + 0 + 2 + 1 + 0.5 + 0.25 = 11.75$$

$$\therefore 1011.11_2 = 11.75_{10}$$

Decimal → Binary (With Fraction) — Successive Multiplication

Multiply the fractional part by the target base. The **integer part of each product** becomes the next digit; carry the remaining fractional part forward. Stop when the fraction hits 0 or you have enough digits.



EXAMPLE: $0.375_{10} \rightarrow \text{BINARY}$

$$0.375 \times 2 = 0.75 \rightarrow \text{integer part } 0, \text{ carry } 0.75$$

$$0.75 \times 2 = 1.5 \rightarrow \text{integer part } 1, \text{ carry } 0.5$$

$$0.5 \times 2 = 1.0 \rightarrow \text{integer part } 1, \text{ carry } 0.0 \text{ (done - fraction is 0)}$$

Read integer parts TOP to BOTTOM: 0, 1, 1

$$\therefore 0.375_{10} = 0.011_2$$

$$\text{Verify: } 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0 + 0.25 + 0.125 = 0.375 \checkmark$$

Octal → Hexadecimal (With Fraction, via Binary)



EXAMPLE: $635.27_8 \rightarrow \text{HEXADECIMAL}$

Step 1: Each octal digit → 3-bit binary (both sides of the point)

$$6=110 \quad 3=011 \quad 5=101 \quad . \quad 2=010 \quad 7=111$$

$$\text{Binary: } 110 \ 011 \ 101 \ . \ 010 \ 111$$

Step 2: Regroup into 4-bit chunks – from the point outward in BOTH directions (left for the whole part, right for the fraction, padding with 0s at each far end)

$$0001 \ 1001 \ 1101 \ . \ 0101 \ 1100$$

$$1 \quad 9 \quad D \quad . \quad 5 \quad C$$

$$\therefore 635.27_8 = 19D.5C_{16}$$

Padding Direction Matters

When regrouping bits around a radix point, pad the **whole-number part on the left** (most significant end) and the **fractional part on the right** (least significant end) — never the opposite, or the value changes.

Binary Arithmetic Operations

Addition Rules

Operation	Result	Carry
0 + 0	0	0
0 + 1	1	0
1 + 0	1	0
1 + 1	0	1 (carry to next column)
1 + 1 + 1 (with carry in)	1	1 (carry to next column)



BINARY ADDITION EXAMPLE

```
  1010  (= 10 decimal)
+ 0111  (=  7 decimal)
-----
 10001  (= 17 decimal)
```

Column by column, right to left:

```
Col 0: 0+1 = 1      → write 1, carry 0
Col 1: 1+1 = 10     → write 0, carry 1
Col 2: 0+1+1(carry) = 10 → write 0, carry 1
Col 3: 1+0+1(carry) = 10 → write 0, carry 1
Col 4: 0+0+1(carry) = 1 → write 1, no carry
```

Result: $10001_2 = 17_{10}$ ✓

Subtraction Rules

Operation	Result	Borrow
$0 - 0$	0	none
$0 - 1$	1	borrow 1 from next column
$1 - 0$	1	none
$1 - 1$	0	none

Why $0 - 1 = 1$?

Just like decimal subtraction, when the top digit is smaller you borrow from the next column to the left. In binary, borrowing 1 adds "2" to the current column (0 becomes $10_2 = 2$, then $2 - 1 = 1$) — that's why the digit result is 1 with a borrow flag set.

Multiplication Rules

Operation	Result
0×0	0
0×1	0
1×0	0
1×1	1

Binary multiplication behaves exactly like decimal long multiplication — multiply by each digit and shift-add the partial products. Since each digit is only 0 or 1, every partial product row is either all-zero or an exact copy of the top number, shifted.

Division Rules

Operation	Result
$0 \div 1$	0

Operation	Result
$1 \div 1$	1
$0 \div 0$	undefined (0 by convention in digit-level tables)
$1 \div 0$	undefined (0 by convention in digit-level tables)

Division by Zero Is Still Undefined

The 0/0 and 1/0 rows appear in the lecture's digit-level reference table for completeness, but real binary long division never actually divides by a 0 digit that would make the operation undefined — the divisor's leading bit is always 1.

Octal & Hexadecimal Arithmetic

Addition, subtraction, multiplication, and division in octal and hexadecimal follow the **exact same carry/borrow logic as decimal arithmetic** — the only difference is that a column "rolls over" to the next place value at 8 (octal) or 16 (hex) instead of at 10.

Base	Operation	Result
Octal	$32_8 + 56_8$	110_8
Octal	$32_8 \times 56_8$	2254_8
Octal	$52_8 \div 25_8$	2_8
Hexadecimal	$32_{16} + 56_{16}$	88_{16}
Hexadecimal	$32_{16} \times 56_{16}$	$10CC_{16}$
Hexadecimal	$52_{16} \div 25_{16}$	2_{16} remainder 8_{16}

Fastest Way to Verify These by Hand

Convert both operands to decimal, do the arithmetic normally, then convert the answer back. E.g. $32_8=26_{10}$ and $56_8=46_{10} \rightarrow 26+46=72_{10} \rightarrow 72_{10}=110_8$.

This is slower on an exam but is the most reliable way to double-check your work while studying.

Filipino Analogy — Binary System

Think of binary like a row of light switches — each switch is either ON (1) or OFF (0). Now imagine 8 switches (1 byte). Each switch's value depends on its position — like how a ₱100 bill is worth more than a ₱1 coin. The leftmost switch in a byte is worth 128, the rightmost is worth 1. The specific pattern of ON/OFF creates any number from 0 to 255.

CHAPTER 05

Data Representation

Every piece of data inside a computer — text, images, audio, video — is ultimately stored as binary numbers (0s and 1s). Data representation is the study of how different types of information are **encoded into binary** and decoded back again.

Bits, Bytes & Data Size Units

Unit	Size	Description & Real-World Example
Bit	1 binary digit (0 or 1)	Smallest unit of digital data. A single transistor state. Like one light switch.
Nibble	4 bits	Half a byte. Represents exactly one hexadecimal digit (0-F).
Byte	8 bits	Smallest addressable memory unit. Stores one ASCII character. "K" = one byte.
Kilobyte (KB)	1,024 bytes	

Unit	Size	Description & Real-World Example
		A short text message or small email. ~1,000 characters.
Megabyte (MB)	1,024 KB	A photo, a short song. One minute of MP3 audio $\approx$ 1 MB.
Gigabyte (GB)	1,024 MB	A movie or hundreds of photos. 1 hour HD video $\approx$ 1-4 GB.
Terabyte (TB)	1,024 GB	Typical external HDD capacity. ~1,000 hours of HD video.
Petabyte (PB)	1,024 TB	Data center scale. Google processes ~20 PB of data per day.

Why 1,024 and not 1,000?

Computer memory uses powers of 2. $2^{10} = 1,024$ — so "kilo" in computing means 1,024, not 1,000. This is why your 256 GB phone shows slightly less storage in Settings — phone manufacturers advertise using decimal (1,000) but the OS measures in binary (1,024). The IEC standard introduced KiB, MiB, GiB (kibibyte, mebibyte, gibibyte) to be precise, but everyday usage still says KB, MB, GB.

ASCII — Text Encoding

ASCII (American Standard Code for Information Interchange) was developed in 1963, primarily by Bob Bemer at IBM. It is a 7-bit encoding standard that maps 128 characters — letters, digits, punctuation, and control characters — to numbers 0-127.

Why it matters: When you type "A" on your keyboard, the computer stores the number 65 (01000001 in binary). ASCII is the translation table between human characters and computer storage.

Range	Characters	Key Values to Know
0-31	Control characters (non-printable)	0 = Null (NUL), 8 = Backspace, 9 = Tab, 10 = Line Feed (newline), 13 = Carriage Return, 27 = Escape
32	Space	Space character = 32. Remember: space is NOT zero.
48-57	Digits '0' through '9'	'0' = 48, '1' = 49, ..., '9' = 57. NOTE: The character '0' is NOT the same as the number 0!
65-90	Uppercase A through Z	'A' = 65, 'B' = 66, ..., 'Z' = 90. (26 uppercase letters)
97-122	Lowercase a through z	'a' = 97, 'b' = 98, ..., 'z' = 122. Always 32 more than the uppercase equivalent.

Key Pattern: Uppercase to Lowercase = +32

'A' = 65. 'a' = 97. Difference = 32. This is how case-conversion works in code. To convert uppercase to lowercase: add 32. To convert lowercase to uppercase: subtract 32. This works for ALL 26 letters of the alphabet.

Unicode — Universal Text Encoding

ASCII only covers 128 characters — it cannot represent Filipino characters (ñ, Á), Chinese, Arabic, emoji, or most of the world's writing systems. **Unicode** was created in 1991 by the Unicode Consortium to solve this permanently.

Property	ASCII	Unicode (UTF-8)
Characters supported	128	1,112,064+ (every language, emoji, mathematical symbols)
Bits per character	7 bits (stored in 8)	Variable: 1-4 bytes per character (UTF-8)

Property	ASCII	Unicode (UTF-8)
Backward compatible?	N/A	YES — UTF-8 first 128 codes match ASCII exactly
Common encodings	One standard	UTF-8 (web standard), UTF-16 (Windows), UTF-32 (fixed-width)
Why UTF-8 dominates	—	Backward compatible, efficient for English text (1 byte), handles all languages, used by 98% of websites

How Images Are Stored

1 Pixel Grid

A digital image is a grid of pixels (picture elements). A 1920×1080 display has $1,920 \times 1,080 = 2,073,600$ pixels. A high-refresh phone display at 2720×1224 resolution has approximately 3.33 million pixels.

2 RGB Color Model

Each pixel stores 3 color channels: Red, Green, Blue. Each channel ranges from 0–255 (8 bits). Combined: 24-bit "True Color" = 16.7 million possible colors. Pure red = R:255, G:0, B:0 stored as `11111111 00000000 00000000` in binary.

3 Color Depth

8-bit = 256 colors. 16-bit = 65,536 colors. 24-bit True Color = 16.7 million colors. 32-bit = 24-bit + 8-bit alpha (transparency channel). Modern AMOLED displays support 10-bit HDR for ~1 billion colors.

4 Image File Formats

BMP — uncompressed, huge files. **JPEG/JPG** — lossy compression, ideal for photos. **PNG** — lossless compression, supports transparency, ideal for graphics/logos. **GIF** — 256 colors, animation support. **WebP** — modern Google format, better compression than JPEG/PNG. **HEIC** — Apple's efficient format for iPhones.

How Audio Is Stored

Sound is an analog wave. To store digitally, it must be **sampled** — measured at regular intervals and converted to numbers via ADC (Analog-to-Digital Converter).

Concept	Description	Standard Values
Sampling Rate	How many times per second the audio wave is measured. Higher = better quality.	44,100 Hz = CD quality; 48,000 Hz = video/ professional standard
Bit Depth	Precision of each measurement sample. Higher = more dynamic range.	16-bit (CD), 24-bit (professional studio recording)
Channels	Number of simultaneous audio streams.	Mono = 1 channel, Stereo = 2 channels, Surround = 5.1/7.1
Audio Formats	File containers/compression methods.	WAV (uncompressed), MP3 (lossy, 128–320 kbps), AAC (Apple, better than MP3), FLAC (lossless), OGG (open- source lossy)

How Video Is Stored

Video = a sequence of still images (frames) displayed rapidly to create motion illusion, synchronized with an audio track.

Concept	Description & Values
Frame Rate	Frames per second (fps). 24 fps = cinema. 30 fps = broadcast TV. 60 fps = smooth motion for gaming/sports. 120/144 fps = high-refresh gaming. Many current gaming-oriented phones support up to a 144Hz display.
Resolution	SD = 480p · HD = 720p · Full HD = 1080p · 4K/UHD = 2160p · 8K = 4320p
Codec	Compression/decompression algorithm. H.264 = most common, compatible everywhere. H.265/HEVC = twice as efficient as H.264. AV1 = open-source, used by YouTube/Netflix. VP9 = Google's format.
Container	File format wrapping video + audio + subtitles. MP4 (universal), MKV (open-source,

Concept	Description & Values
	multiple tracks), AVI (legacy Windows), MOV (Apple).

Data Compression

Type	Description	Examples	Best Used For
Lossless	Original data perfectly recovered after decompression. Removes redundancy without discarding any data.	ZIP, PNG, FLAC, GIF (animation), LZ4	Text documents, software, executable files, medical images — anywhere exact data recovery is required.
Lossy	Some data permanently discarded. Achieves much higher compression by exploiting human perception limits (eyes/ears can't notice small losses).	JPEG, MP3, H.264/H.265, AAC	Photos, music, video where small quality loss is acceptable in exchange for much smaller file sizes.

CHAPTER 06

Operating Systems

An Operating System (OS) is the **master software layer** that manages all hardware resources and provides a platform for application software to run. Without an OS, application software cannot function. It is the bridge between the user, software, and hardware.

Filipino Analogy — The Operating System

Think of the OS as the **Barangay Captain**. It manages all community resources (hardware), lets residents (applications) use shared facilities (CPU, memory, printer), settles conflicts when two residents want the same resource at the same time (process scheduling), maintains records (file system), and enforces rules (security/permissions). Without a Barangay Captain, everyone fights over resources and nothing works in order.

Core Functions of an Operating System

Function	What It Manages	How It Works
Process Management	CPU time, running programs	Creates, schedules, and terminates processes. Handles multitasking via time-slicing — each process gets tiny CPU timeslots in rapid rotation. Manages process priority queues and prevents deadlocks.
Memory Management	RAM allocation	Allocates and deallocates RAM to processes. Implements virtual memory (uses SSD/HDD space as overflow RAM via paging). Prevents processes from accessing each other's memory (memory protection).
File System Management	Files and directories	Organizes data in hierarchical folder/directory structures. Manages read, write, delete, rename operations. Handles file permissions (who can access what). Maintains file metadata.
Device Management	Hardware peripherals	Uses device drivers to communicate with hardware. Manages I/O queues. Handles plug-and-play

Function	What It Manages	How It Works
System Management	File system, disk space, hardware resources	Manages file system (directories, files, permissions). Manages disk space (partitioning, formatting, file allocation). Manages hardware resources (CPU, memory, I/O).
	Process management, system services	Manages processes (creation, execution, termination). Manages system services (daemons, background processes).
Security & Access Control	Authentication, permissions	Verifies user login credentials. Manages file permissions (read/write/execute for owner/group/others). Enforces isolation between user accounts. Logs security events.
User Interface	Human-computer interaction	GUI (Graphical User Interface) — icons, windows, menus, mouse. CLI (Command-Line Interface) — text commands in a terminal. Your Debian setup uses both (XFCE4 GUI + bash terminal).
Error Handling	System faults, crashes	Handles hardware failures and application crashes gracefully. Generates error logs and crash dumps. Prevents one failing program from crashing the entire system.

Types of Operating Systems

Type	Description	Examples
Batch OS	Processes jobs in batches without real-time user interaction. Jobs are queued and run sequentially without user involvement during execution. No direct user-computer interaction while processing.	Early IBM mainframes, IBSYS. Used for payroll processing, billing runs.
Time-Sharing OS (Multi-user)	Multiple users share CPU time simultaneously. CPU rapidly switches between	UNIX, Multics, modern Linux servers hosting multiple user accounts.

Type	Description	Examples
	users in small timeslices, giving each user the illusion of exclusive access.	
Real-Time OS (RTOS)	Processes inputs and responds within strict, guaranteed time constraints. Hard RTOS: missing a deadline = system failure (airplane autopilot, pacemaker). Soft RTOS: occasionally missing is acceptable (video streaming).	FreeRTOS, VxWorks, QNX, embedded Linux in IoT devices, medical equipment, automotive.
Distributed OS	Manages a network of computers as a single unified system. Resources distributed across multiple machines appear as one system to the user.	Google's Borg cluster manager, Kubernetes, Hadoop YARN.
Network OS	Designed specifically to manage network resources: shared file servers, printers, user authentication from remote computers.	Windows Server, Linux Server (Debian/Ubuntu), Novell NetWare.
Mobile OS	Optimized for touch interfaces, battery efficiency, wireless connectivity, app stores, sensors (GPS, accelerometer, camera).	Android (Linux-based, Google/AOSP), iOS (XNU kernel, Apple), HarmonyOS (Huawei).
Embedded OS	Tiny OS running inside devices with very limited resources — smart TVs, routers, washing machines, cars, IoT sensors, ATMs.	Linux (Raspberry Pi), FreeRTOS, Contiki, ThreadX, VxWorks.

| Popular Operating Systems — Deep Look

Windows (Microsoft)

- **Creator:** Microsoft (Bill Gates & Paul Allen)
- **First release:** Windows 1.0 — November 20, 1985 (GUI shell on top of MS-DOS)
- **Kernel:** Windows NT kernel (since Windows NT 3.1, 1993). 64-bit architecture.
- **Major versions:** Windows 95 (huge milestone), XP (most loved), Vista (failure), 7 (great), 8 (failure), 10 (most used), 11 (current)
- **Market share:** ~73% of desktop OS worldwide
- **File system:** NTFS (primary), exFAT (removable drives), FAT32 (legacy USB)
- **Strengths:** Massive software compatibility, gaming, enterprise support, Microsoft Office
- **Weaknesses:** Higher malware risk, heavier resource usage, licensing cost

Linux

- **Creator:** Linus Torvalds — first released September 17, 1991 as a free, open-source kernel
- **What it is:** A kernel, not a full OS. A Linux distribution (distro) = kernel + GNU tools + package manager + desktop environment
- **Example setup:** Debian Trixie (Linux distro) + XFCE4 (desktop environment) running inside a Linux-on-Android compatibility layer on a mid-range smartphone
- **Major distros:** Ubuntu (beginner-friendly), Debian (rock-stable — what you use), Fedora (cutting-edge), Arch Linux (advanced DIY), Kali (security testing), Red Hat/RHEL (enterprise)
- **File system:** ext4 (primary default), btrfs (modern), XFS (high-performance)
- **Market share:** ~2% desktop, 96% web servers, 100% of top 500 supercomputers, all Android phones
- **Strengths:** Free, open-source, highly customizable, extremely secure, runs everywhere

macOS (Apple)

- **Creator:** Apple Inc. (Steve Jobs, Tim Cook)
- **First release:** Mac OS X 10.0 "Cheetah" — March 24, 2001
- **Kernel:** XNU (hybrid kernel combining Mach microkernel + BSD UNIX)
- **Hardware:** Exclusive to Apple hardware (Mac, MacBook, iMac, Mac Pro)

- **Strengths:** Rock-solid stability, excellent for creative work, tight hardware-software integration, great build quality

Android

- **Creator:** Andy Rubin (Android Inc.), acquired by Google in 2005
- **First release:** Android 1.0 — September 23, 2008 (HTC Dream)
- **Foundation:** Linux kernel + Android Runtime (ART) + Java/Kotlin apps
- **Market share:** ~72% of global smartphones
- **Example phone:** A mid-range Android phone running a manufacturer custom skin (based on Android 15) on Linux kernel 6.x

The Booting Process

Boot = the complete sequence of events from pressing the power button to seeing the login screen.

1 Power ON

Electrical power flows to all components. Capacitors stabilize voltage. CPU begins execution from a fixed, hardwired address in ROM where the BIOS/UEFI firmware lives.

2 POST — Power-On Self-Test

BIOS/UEFI performs hardware health checks: RAM quantity and functionality, CPU, keyboard presence, storage devices, GPU. If something critical fails: beep codes (one beep usually = success). This is why PCs sometimes beep on startup.

3 BIOS / UEFI Initializes

BIOS (Basic Input/Output System) is older firmware, 16-bit, max 2.2 TB drives.

UEFI (Unified Extensible Firmware Interface) is modern: 32/64-bit, supports drives >2.2 TB, faster boot (no 1-minute BIOS logo screens), Secure Boot. Both live in a Flash ROM chip on the motherboard.

4 Boot Device Found

BIOS/UEFI checks the boot order list (SSD first, then USB, then network...) and finds the bootable device. Reads the MBR (Master Boot Record, legacy) or ESP (EFI System Partition, UEFI) to locate the OS bootloader.

5 Bootloader Runs

Windows: Windows Boot Manager. Linux: **GRUB** (Grand Unified Bootloader). The bootloader knows where the OS kernel files are and loads them into RAM. If dual-booting (both Windows and Linux), GRUB shows a menu to choose.

6 OS Kernel Loads

The kernel loads into RAM and takes over from the bootloader. Initializes all hardware drivers, mounts the root file system, sets up memory management, starts the first process.

7 Init System Starts Services

Linux uses **systemd** (modern) or **init** (legacy) to start all background services: networking, display manager, audio daemon, bluetooth, login manager. Windows uses Services Manager and Session Manager.

8 Login Screen Appears

The graphical display manager (LightDM, GDM, SDDM on Linux; Winlogon on Windows) presents the login screen. User authenticates and the desktop environment loads.

File Systems

File System	OS	Max File Size	Key Features
FAT32	Universal	4 GB limit!	Oldest, most compatible. Used on USB drives. Cannot store files >4 GB — this is why you can't copy a large video file to a FAT32 USB drive.
exFAT	Windows, Mac, Android	16 EB	"FAT32 upgraded" — no 4 GB limit, cross-platform. Ideal for large USB drives and SD cards used with multiple OS types.
NTFS	Windows	16 EB	Windows default since NT. Journaling (recovers from crashes), file permissions, BitLocker encryption support, file compression.
ext4	Linux	16 TB	Linux default. Journaling, very

File System	OS	Max File Size	Key Features
			stable, supports UNIX permissions. Your Debian partition uses ext4.
APFS	macOS, iOS	8 EB	Apple's modern format. Copy-on-write, snapshots, built-in encryption, space sharing between volumes.

CHAPTER 07

Software Classification & Programming Languages

Software is the set of **instructions** that tell hardware what to do. Where hardware is physical, software is logical — code, programs, and data. This chapter classifies all software types and traces how human-written code becomes machine-executable instructions.

| System Software vs Application Software

SOFTWARE

- └─ System Software (manages the computer itself)
 - | └─ Operating System – Windows, Linux, macOS, Android
 - | └─ Utility Programs – antivirus, disk cleanup, backup tools, file compressors
 - | └─ Device Drivers – printer driver, GPU driver, WiFi adapter driver
 - | └─ Language Translators – compilers, interpreters, assemblers
 - |
- └─ Application Software (helps the USER do tasks)
 - | └─ General Purpose – MS Word, Google Chrome, Spotify, Calculator

- └ Specialized / Special Purpose – Photoshop, AutoCAD, SAP, MATLAB
- └ Custom / Bespoke Software – company-built internal systems

Filipino Analogy — System vs Application Software

System Software is like the **MMDA** — it manages and maintains the road infrastructure itself (traffic lights, signage, road surface) so that traffic can flow at all. Application Software is like **Grab or Angkas** — it uses that existing road infrastructure to deliver a specific service to the end user. You never deal with MMDA directly day-to-day, but without it, Grab couldn't operate.

System Software Breakdown

Category	Function	Examples
Operating System	Manages all hardware resources, provides platform for apps to run	Windows 11, Debian Linux, macOS, Android, iOS
Utility Programs	Maintain, optimize, and protect the system	Antivirus (Windows Defender), Disk Cleanup, WinRAR/7-Zip, CCleaner, Backup tools
Device Drivers	Translate OS commands into hardware-specific signals	NVIDIA GPU driver, HP printer driver, Realtek audio driver, MediaTek chipset driver
Language Translators	Convert human-written code into machine-executable instructions	javac (Java compiler), gcc (C compiler), Python interpreter, NASM (assembler)

Application Software Breakdown

Category	Description	Examples
General Purpose	Broad everyday use across many tasks and users	MS Word, Excel, Chrome, Calculator, VLC Media Player
Specialized / Special Purpose	Built for a specific professional task or industry	AutoCAD (engineering), Photoshop (design), SAP (enterprise resource

Category	Description	Examples
		planning), MATLAB (engineering math)
Custom / Bespoke Software	Built specifically for one company's unique workflow — not sold publicly	A bank's internal loan processing system, QCU's own student portal/registrar system

Application Software by Function

Application software can also be classified by the specific task it performs. This is the breakdown most often tested alongside the Service Learning module (Chapter 11):

Application Software Type	Examples
Word Processing Software	MS Word, WordPad, Notepad
Database Software	Oracle, MS Access
Spreadsheet Software	Microsoft Excel, Apple Numbers
Multimedia Software	Real Player, Media Player
Presentation Software	Microsoft PowerPoint, Keynote, Canva
Enterprise Software	Customer relationship management (CRM) systems

Generations of Programming Languages

Programming languages evolved across five generations, each moving **further from raw hardware instructions and closer to natural human language.**

Gen	Name	Description	Examples
1GL	Machine Language	Pure binary (0s and 1s). Directly executed by CPU. Extremely fast but nearly impossible	10110000 01100001

Gen	Name	Description	Examples
		for humans to read or write.	
2GL	Assembly Language	Uses mnemonic codes (short human-readable abbreviations) representing machine instructions. Requires an Assembler to convert to machine code. Still hardware-specific.	MOV AX, 5 / ADD BX, AX
3GL	High-Level Languages	Human-readable syntax resembling English/math. Requires a Compiler or Interpreter. Portable across hardware. This is where Java (CC 102) lives.	Java, C, C++, Python, COBOL, FORTRAN
4GL	Very High-Level / Declarative	Specifies WHAT result you want, not HOW to get it step-by-step. Often domain-specific (databases, reporting).	SQL, MATLAB, R
5GL	AI / Constraint-Based	Uses logical constraints or natural language to solve problems; the system figures out the steps. Used in AI research.	Prolog, OPS5, modern AI prompt-based coding

Compilers vs Interpreters

Both are **Language Translators** — they convert high-level source code into something the CPU can execute. They differ fundamentally in HOW and WHEN that translation happens.

Aspect	Compiler	Interpreter
Translation timing	Translates the ENTIRE program at once, before execution	Translates and executes ONE line at a time, during execution
Output	Produces a separate executable file (.exe, .class)	No separate file — runs directly from source each time
Speed	Faster execution (already translated)	Slower execution (translates on the fly, every run)
Error detection	Reports ALL errors after scanning the whole program	Stops at the FIRST error it encounters
Examples	gcc (C/C++), javac (Java)	Python (CPython), original JavaScript engines, PHP

Forward Link — CC 102: Java Is a HYBRID

Java doesn't fit neatly into either box — it uses BOTH. The **javac** compiler first compiles your **.java** source file into platform-independent **bytecode** (**.class** file). Then the **JVM (Java Virtual Machine)** interprets — or JIT (Just-In-Time) compiles — that bytecode into native machine code at runtime. This is exactly why Java's slogan is "Write Once, Run Anywhere" — the bytecode runs on any device with a JVM installed, including a Linux-on-Android dev environment running Debian/JDK 17. You'll experience this directly on Day 1 of CC 102 when you type **javac Hello.java** then **java Hello** — two separate steps because compile and run are distinct phases.

Software Licensing Models

Model	Description	Examples
Proprietary / Commercial		

Model	Description	Examples
	Closed source code, must be purchased or licensed. Cannot legally view, modify, or redistribute the code.	Windows, MS Office, Adobe Photoshop, AutoCAD
Open Source / FOSS	Source code is publicly available to view, modify, and redistribute, usually for free, governed by a license (GPL, MIT, Apache).	Linux, Firefox, LibreOffice, VS Code (source-available), the Java OpenJDK you use
Freeware	Free to use, but source code remains closed/proprietary. No payment required, but no modification rights.	Older Skype, Adobe Acrobat Reader, Zoom (basic tier)
Shareware	Free trial period or limited features, requires payment to unlock full functionality.	WinRAR (technically "free forever" trial), many mobile games with paywalls

CHAPTER 08

Basic Networking & the Internet

A computer network is a group of two or more devices connected to share data and resources. This chapter covers the foundations you need before **NET 101** (Networking 1) goes deep in Year 2, and gives you the mental model for understanding how **WS 101** web requests actually travel across the internet.

Types of Networks by Geographic Scope

Type	Full Name	Coverage	Example
PAN	Personal Area Network	A few meters	Bluetooth connection between your phone and earbuds

Type	Full Name	Coverage	Example
LAN	Local Area Network	Single building/home	Your home WiFi connecting your laptop, phone, and smart TV
MAN	Metropolitan Area Network	A city	QCU connecting all its campus buildings together
WAN	Wide Area Network	Country/Continent	PLDT or Globe's nationwide network; the Internet itself is the largest WAN

Network Topologies

Topology = the physical or logical arrangement of how devices connect to each other.

Topology	Structure	Pros	Cons
Star	All devices connect to one central hub/switch	Easy to manage, one device failing doesn't affect others	If the central hub fails, the entire network goes down
Bus	All devices share a single central cable	Cheap, simple to install	If the main cable fails, the entire network goes down; slows with more devices
Ring	Devices connected in a circular chain, data passes around the ring	Equal access for all devices	One broken connection can disrupt the entire ring
Mesh	Every device connects directly to every other device	Extremely reliable — many redundant paths	Expensive, complex cabling, hard to scale
Tree	Hierarchical combination of star networks	Scalable, organized (good for large institutions like QCU)	If a parent node fails, all its children lose connection

Topology	Structure	Pros	Cons
Hybrid	Combination of two or more different topologies (e.g. star + ring), wired or wireless	Lets an administrator balance stability and bandwidth as needed for different parts of the network	More complex to design, install, and manage than a single topology

IP Addresses

An **IP Address** (Internet Protocol Address) is a unique numerical label assigned to every device on a network, allowing it to be located and communicated with — similar to how a home address lets mail find you.

Property	IPv4	IPv6
Bit length	32-bit	128-bit
Format	Dotted decimal: 192.168.1.1	Hexadecimal colon-separated: 2001:0db8::ff00:42
Total addresses	~4.3 billion	340 undecillion (effectively unlimited)
Status	Exhausted globally as of 2019 — this is WHY IPv6 exists	Gradual global rollout, ~45% adoption as of recent years

Public vs Private IP Addresses

- **Private IP:** Used inside a local network only (home/office). Common ranges: **192.168.x.x**, **10.x.x.x**. Your laptop's IP at home is private — invisible to the outside internet directly.
- **Public IP:** Globally unique, assigned by your ISP (Globe/PLDT/Converge), visible to the entire internet. Your home router has ONE public IP that all your devices share via NAT (Network Address Translation).

IP Address vs MAC Address

	IP Address	MAC Address
Assigned by	Network/router (can change)	Hardware manufacturer (burned into the network card, permanent)
Format	192.168.1.1 (IPv4)	00:1A:2B:3C:4D:5E (48-bit hex)
Scope	Logical address, identifies network location	Physical address, identifies the specific hardware device
Layer	Network Layer (OSI Layer 3)	Data Link Layer (OSI Layer 2)

DNS — Domain Name System

Humans remember names; computers route by numbers. DNS is the internet's "phonebook" — it **translates human-friendly domain names into IP addresses**.

1 You type a URL

e.g. `www.qcu.edu.ph` into your browser.

2 Browser asks a DNS Resolver

Usually your ISP's DNS server, or a public one like Google (8.8.8.8) or Cloudflare (1.1.1.1).

3 DNS lookup chain

Resolver queries Root servers → TLD servers (.ph, .com) → Authoritative server for that exact domain, until it finds the IP.

4 IP address returned

e.g. `203.177.x.x` is sent back to your browser.

5 Connection established

Browser now connects directly to that IP address to load the actual website.

HTTP vs HTTPS

	HTTP	HTTPS
Full name	HyperText Transfer Protocol	HTTP Secure
Encryption	None — data sent in plain text, readable if intercepted	TLS/SSL encryption — data is scrambled, unreadable if intercepted
Port	80	443
Browser indicator	"Not Secure" warning	Padlock icon
Use today	Essentially deprecated for any sensitive data	Standard for virtually all modern websites, including GCash, online banking

Forward Link — WS 101: HTTP in Practice

In WS 101 you'll build pages that get requested via HTTP/HTTPS. Every time a browser loads your HTML file, it's making an HTTP GET request to a server and receiving the file back as the response — exactly the client-server model you're learning right now in CC 101.

The OSI Model — 7 Layers

The OSI (Open Systems Interconnection) Model is a conceptual framework describing how data travels through a network, broken into 7 distinct layers. Each layer has a specific job and only talks to the layers directly above and below it.

7 Application

HTTP, FTP, SMTP, DNS

Closest to the user — what apps/browsers use directly

6 Presentation

SSL/TLS, encryption

Formats/encrypts data — translation, compression, encryption

5	Session	NetBIOS, RPC	Opens, manages, and closes connections between two devices
4	Transport	TCP, UDP	Ensures complete data delivery, breaks data into segments
3	Network	IP, ICMP	Handles addressing and routing — chooses the best path
2	Data Link	Ethernet, MAC, WiFi (802.11)	Handles physical addressing (MAC) within the same local network
1	Physical	Cables, radio waves, voltage	The actual physical transmission medium — electrical signals, light pulses, radio

OSI Memory Trick (top to bottom)

"**A**ll **P**eople **S**ee**m** **T**o **N**eed **D**ata **P**rocessing" — Application, Presentation, Session, Transport, Network, Data Link, Physical. This 7-layer model is foundational and you'll see it again formally in **NET 101** (Year 2, Sem 1).

Bandwidth, Latency & Throughput

Term	Definition	Analogy
Bandwidth	Maximum theoretical data capacity of a connection, measured in Mbps/Gbps	Width of a highway — how many lanes total
Latency	Time delay for data to travel from source to destination, measured in milliseconds (ms)	How long it takes one car to drive the highway's full length
Throughput	Actual real-world data transfer rate achieved (always $\leq$ bandwidth)	Actual number of cars that get through per minute, accounting for traffic
Ping	Round-trip latency test — how long for a signal to go to a destination and back	Important in online gaming (Mobile Legends, Wuthering Waves multiplayer) — high ping = noticeable lag

Network Devices

Device	OSI Layer	Function
Hub	Physical (1)	Broadcasts incoming data to ALL connected ports — outdated, inefficient, rarely used today
Switch	Data Link (2)	Sends data only to the specific intended device using MAC addresses — much smarter than a hub
Router	Network (3)	Connects different networks together (e.g. your home LAN to the wider Internet); assigns private IPs, performs NAT
Modem	Physical (1)	Converts digital signals to/ from the analog signals used

Device	OSI Layer	Function
		by ISP infrastructure (fiber, DSL, cable)
Access Point	Data Link (2)	Extends a wired network to provide wireless (WiFi) connectivity
Firewall	Network/Transport	Filters incoming/outgoing traffic based on security rules — can be hardware or software

Internet vs Web vs Intranet — Don't Confuse These

Term	What It Actually Is
Internet	The massive global physical network of interconnected computers and cables/satellites — the infrastructure itself.
World Wide Web (WWW)	A SERVICE that runs on top of the Internet — the collection of websites and pages accessed via HTTP/HTTPS. Invented by Tim Berners-Lee in 1989. The Internet existed before the Web.
Intranet	A PRIVATE network using internet technology, restricted to one organization. QCU's internal student portal/grading system is an intranet — accessible only to QCU's network/credentials.
Extranet	A controlled private network extended to selected outside parties (e.g. QCU sharing limited portal access with CHED for scholarship verification).

Advantages & Disadvantages of Networking

UPSIDE Advantages	TRADE-OFF Disadvantages
-----------------------------	-----------------------------------

Enhances communication and availability of information; enables convenient resource and file sharing; highly flexible; increases cost efficiency by sharing hardware; boosts overall storage capacity.

Reduces device independence; introduces security difficulties (more entry points for attackers); lacks robustness (one failure can affect many users); increases exposure to viruses/malware; requires a skilled administrator and, often, an expensive set-up.

Network Configuration: Peer-to-Peer vs. Server-Based

Beyond geographic scope, networks are also classified by how devices relate to each other administratively.

	Peer-to-Peer (Workgroup Model)	Server-Based (Domain / Client-Server Model)
Structure	Every computer acts as both a workstation and a server	Dedicated server(s) hold shared resources; other computers act as clients
Management	Each computer is managed independently, with equal rights to initiate communication, share resources, and validate users	Centralized — the server controls the level of access each client has
Best for	Small setups, e.g. a handful of home/office PCs	Larger organizations — shared data lives in one location, making backups easier

Types of Wired Network Media

Media	How It Carries Data	Notes
Twisted Pair	Electronic signals through two insulated copper wires twisted around each other	UTP (Unshielded) has no shielding; STP (Shielded) adds shielding to reduce electromagnetic interference

Media	How It Carries Data	Notes
		(EMI). Typical speed 10-100 Mbps, max length ~100m.
Coaxial Cable ("coax")	Electrical signals through a center wire, insulation, and a grounded braided-wire shield	More resistant to interference than basic twisted pair; commonly used for cable internet/TV drops.
Fiber Optic	Modulated pulses of light through light-conducting glass or plastic fibers	Fastest and most interference-resistant wired medium; core fiber is protected by cladding and a tough outer sheath.

UTP Cable Configurations

- **Straight-Through Cable** — the most common configuration; connects dissimilar devices, e.g. a computer to a hub or switch.
- **Crossover Cable** — connects two devices of the same type directly, e.g. computer to computer, or switch to switch.

More Network Devices

Beyond Hub, Switch, Router, Modem, Access Point, and Firewall covered above, three more devices come up regularly:

Device	Function
Repeater	Retransmits a received signal with more power, extending a network's geographical or topological reach beyond what the original signal could cover alone.
Bridge	Interconnects two network segments that use the same protocol — e.g. joining a Corporate LAN segment with a Manufacturing segment.
Gateway	Provides a network with connectivity to a remote network or autonomous system that is otherwise out of bounds for the host network's own nodes.

Additional Network Terminology

Term	Meaning
Backbone	The main high-capacity cable/connection that other network segments branch off from.
Packet	A formatted unit of data carried by a network — a message is broken into packets, sent, then reassembled at the destination.
PING (Packet Internet Groper)	A utility that tests whether a host is reachable by sending a small packet and measuring the response.
Port	A numbered logical endpoint on a device that identifies which application/service a piece of network traffic belongs to (e.g. Port 80 for HTTP).
Route	The path data takes from source to destination across a network.
Subnet / Subnetting / Subnet Mask	A subnet is a smaller network carved out of a larger one; subnetting is the process of dividing it; a subnet mask marks which part of an IP address is the network portion versus the host portion.
T1 Line	A dedicated, high-speed digital telecommunications line historically used for business-grade internet/data connections.
UNC (Universal Naming Convention)	A standard format for specifying the location of a shared resource on a network, e.g. \\server\folder.
VPN (Virtual Private Network)	Creates an encrypted, private tunnel over a public network so a remote user can securely access a private network as if directly connected to it.
Collision	Occurs when two devices transmit data on a shared medium at the same time, corrupting both signals.

Information Systems

An Information System (IS) combines people, hardware, software, data, and processes to collect, process, store, and distribute information that supports decision-making within an organization. This chapter introduces the IS types tested in CC 101 and previews concepts that return in full depth in **CC 105** (Information Management, Year 2) and **IS 106** (IS Project Management, Year 2).

Data vs Information vs Knowledge

Term	Definition	Example
Data	Raw, unprocessed facts with no context on their own	"85", "92", "78" — just numbers
Information	Data that has been processed, organized, and given context/meaning	"Kian scored 85 in CC 101, 92 in WS 101, 78 in CC 102"
Knowledge	Information combined with experience, enabling understanding and decision-making	"Kian is strongest in WS 101 — should consider focusing extra practice on CC 102"

The 5 Components of an Information System

Component	Role
Hardware	Physical devices — servers, computers, networking equipment that run the system
Software	Programs and applications that process the data — the actual IS application
Data	The raw facts being collected, stored, and processed by the system

Component	Role
People	Users, IT staff, and decision-makers who interact with and maintain the system
Process	The defined procedures and rules for how data flows through the system

Types of Information Systems by Organizational Level

Organizations have layered levels of management — Information Systems are designed to serve each level's specific decision-making needs.

OPERATIONAL LEVEL

TPS — Transaction Processing System

Records and processes the day-to-day routine transactions essential for business operations. High volume, repetitive, structured data.

Example: ATM withdrawals, POS checkout at SM/7-Eleven, payroll processing, QCU enrollment system recording each student's registration

TACTICAL / MIDDLE MANAGEMENT

MIS — Management Information System

Summarizes TPS data into structured reports for middle managers to monitor performance and make routine decisions.

Example: Monthly sales summary report, weekly attendance report for a department head

TACTICAL / ANALYTICAL

DSS — Decision Support System

Provides analytical tools (what-if analysis, forecasting, modeling) to support semi-structured or unstructured decisions.

Example: A logistics company simulating different delivery routes to minimize fuel cost

STRATEGIC / EXECUTIVE LEVEL

ESS — Executive Support System

Gives top executives high-level dashboards and summaries for long-term strategic planning, pulling from internal AND external data sources.

Example: A CEO's dashboard showing company-wide KPIs, market trends, and competitor analysis

Filipino Analogy — IS Organizational Pyramid

Think of a Barangay structure: **TPS** = the Barangay Tanod logging daily incident reports (raw operational data). **MIS** = the Barangay Secretary compiling those into a weekly summary for the Kagawad. **DSS** = the Kagawad running "what-if" scenarios on budget allocation. **ESS** = the Barangay Captain's big-picture dashboard combining all of this plus city-wide data to set long-term Barangay policy.

Other Notable Information System Types

Type	Description	Example
Expert System (ES)	Mimics human expert decision-making in a specific narrow domain using rule-based logic ("if-then" reasoning)	Medical diagnosis support systems, tax preparation software
Office Automation System (OAS)	Improves productivity of office workers through document management and communication tools	MS Office 365, Google Workspace, Slack, email systems
Knowledge Management System (KMS)	Captures, stores, and shares organizational knowledge so it isn't lost when employees leave	Company wikis, internal documentation portals, Confluence

Intro to Databases — Foundation Concepts

Information Systems need somewhere to store data persistently — that's a **database**. CC 101 only requires the basic vocabulary; full database design comes in **CC 105** and **IM 101** in Year 2.

Term	Definition
Database	An organized collection of structured data stored electronically
Table	A structured set of data organized in rows and columns, like a spreadsheet

Term	Definition
Record (Row)	A single complete entry in a table — e.g., one student's full record
Field (Column)	A single category of data within a record — e.g., "Student Name," "Course," "Year Level"
Primary Key	A unique identifier for each record so no two records are confused — e.g., your Student No. 12-3456
DBMS	Database Management System — software that creates and manages databases (MySQL, PostgreSQL, Oracle, SQL Server, MongoDB)

CHAPTER 10

Computer Security & Ethics

This chapter covers the threats facing computer users, the basic defenses against them, the Philippine laws governing computer use, and the ethical principles every IT professional is expected to follow. This lays the groundwork for the full dedicated course **IAS 101** (Fundamentals of Information Assurance & Security, Year 3 Sem 2).

| Types of Malware

Malware (malicious software) is any software intentionally designed to cause damage, steal data, or gain unauthorized access.

Virus Attaches itself to a legitimate file or program. Requires a host file and human action (opening the infected file) to activate and spread. Can corrupt or delete data.

Worm Self-replicating malware that spreads automatically across networks WITHOUT needing a host file or human action — exploits network vulnerabilities directly. Can spread far faster than viruses.

Trojan Horse

Disguises itself as legitimate, useful software to trick users into installing it. Named after the Greek mythological story. Once installed, it can open backdoors for attackers.

Ransomware	Encrypts the victim's files, making them inaccessible, then demands payment (usually cryptocurrency) for the decryption key. Major real-world example: WannaCry (2017) affected 200,000+ computers across 150 countries.
Spyware	Secretly monitors user activity — keystrokes, browsing habits, screen content — and sends data to an attacker without consent.
Adware	Automatically displays unwanted advertisements, often bundled with free software. Less destructive than other malware but invasive and can slow systems significantly.
Rootkit	Designed to hide deep within the OS, granting an attacker persistent privileged (admin/root) access while remaining undetected by normal antivirus scans.
Keylogger	Records every keystroke typed by the user, capturing passwords, messages, and sensitive data — often a component within spyware or trojans.
Privilege Escalation	A technique where an attacker gains access rights beyond what was originally granted, often by exploiting a bug or misconfiguration, to reach admin/root-level control.
Botnet	A network of malware-infected computers ("bots") remotely controlled by an attacker, commonly used to launch large-scale attacks such as DDoS or to send spam.
Logic Bomb	Malicious code deliberately inserted into software that stays dormant until triggered by a specific condition or date, at which point it executes its damaging payload.
Spam	Unsolicited, often bulk-sent messages — more of a nuisance than direct malware on its own, but a common delivery vector for phishing links and other malware.

Cybersecurity Defense Basics

Defense	How It Works
Strong Passwords	Long (12+ characters), mixed case, numbers, symbols, unique per account. Avoid dictionary words and personal info (birthdays, names).
Multi-Factor Authentication (MFA/2FA)	Requires 2+ proof types: something you KNOW (password), something you HAVE (phone OTP), something you ARE (fingerprint/face). Even if a password leaks, the account stays protected.
Encryption	Scrambles data using a mathematical algorithm so only someone with the correct key can read it. Symmetric: same key encrypts and decrypts (fast, used for bulk data). Asymmetric: public key encrypts, private key decrypts (used in HTTPS, digital signatures).
Firewall	Monitors and filters incoming/outgoing network traffic based on security rules, blocking unauthorized access attempts.
Antivirus / Anti-malware	Scans files and running processes against known malware signatures and suspicious behavior patterns, then quarantines or removes threats.
Regular Backups	Maintaining separate copies of important data so ransomware or hardware failure doesn't mean permanent data loss.
Software Updates / Patches	Regularly applying security patches closes known vulnerabilities before attackers can exploit them.

| Phishing & Social Engineering

Social engineering exploits human psychology rather than technical vulnerabilities — tricking people into giving up information or access voluntarily.

Type	Description
Phishing	Fraudulent emails/messages impersonating a trusted entity (bank, GCash, school) to trick victims into clicking malicious links or revealing credentials.
Smishing	Phishing via SMS text message — very common in the Philippines (fake "GCash account locked" texts, fake prize notifications).
Vishing	Phishing via voice call — fake bank representative calling to "verify" your account details.
Pretexting	Attacker invents a fabricated scenario/identity to gain trust and extract information.

| Philippine Cyber & Data Laws

REPUBLIC ACT 10175

Cybercrime Prevention Act of 2012

Signed into law: September 12, 2012

The primary law criminalizing offenses committed via computer systems and the internet in the Philippines. Covers: illegal access (hacking), data interference, cyber-squatting, computer-related fraud and identity theft, cybersex, child pornography online, and online libel. Established the creation of the Cybercrime Investigation and Coordinating Center.

REPUBLIC ACT 10173

Data Privacy Act of 2012

Signed into law: August 15, 2012

Protects individual personal information stored in both government and private sector systems. Established the National Privacy Commission (NPC) as the regulating body. Mandates organizations implement reasonable security measures and notify affected individuals/NPC of data breaches.

REPUBLIC ACT 8792

Electronic Commerce Act of 2000

Signed into law: June 14, 2000

Gives legal recognition to electronic documents, electronic signatures, and electronic transactions — making them as legally valid and enforceable as paper-based ones. Foundation law for all e-commerce and online contracts in the Philippines.

REPUBLIC ACT 8293

Intellectual Property Code of the Philippines

Signed into law: June 6, 1997

Protects copyrights, trademarks, and patents — including software and source code as intellectual property. Relevant to you as a future developer: your original code, apps, and projects are automatically protected under this law.

REPUBLIC ACT 11934

SIM Registration Act

Signed into law: October 10, 2022

Requires all SIM card users in the Philippines to register their personal information with telecom providers, intended to curb text scams and SMS-based fraud — directly related to the smishing threats covered above.

The Ten Commandments of Computer Ethics

Published by the Computer Ethics Institute (Washington D.C.) in 1992 — still the most widely cited ethical framework in computing education worldwide, and a recurring CC 101 exam topic.

1. **Thou shalt not use a computer to harm other people.**
2. **Thou shalt not interfere with other people's computer work.**
3. **Thou shalt not snoop around in other people's computer files.**
4. **Thou shalt not use a computer to steal.**

5. **Thou shalt not use a computer to bear false witness.**
6. **Thou shalt not copy or use proprietary software for which you have not paid.**
7. **Thou shalt not use other people's computer resources without authorization.**
8. **Thou shalt not appropriate other people's intellectual output.**
9. **Thou shalt think about the social consequences of the program you write.**
10. **Thou shalt always use a computer in ways that show consideration and respect.**

| Netiquette — Online Etiquette Basics

- **Think before you post** — once online, content can be screenshotted and spread permanently, even if deleted.
- **No ALL CAPS** — widely interpreted as shouting in digital communication.
- **Respect privacy** — don't share others' personal information, photos, or messages without consent.
- **Cite sources** — give credit when sharing others' work, ideas, or content.
- **Stay on topic** — avoid derailing group chats, forums, or class discussion threads.
- **Verify before sharing** — combat misinformation by checking facts before resharing news or claims.

| The Three Goals of Computer Security — CIA Triad

Every computer security effort ultimately serves one of three goals, commonly remembered as the **CIA Triad**.

Goal	Definition	Common Tools
Confidentiality	Ensures that computer-related assets are accessed only by authorized parties.	Encryption, Access Control, Authentication, Authorization, Physical Security
Integrity	Ensures that assets can be modified only by authorized parties, or only in authorized ways.	Backups, Checksums, Data Correcting Codes
Availability	Ensures that assets are accessible to authorized parties at appropriate times.	Physical Protections, Computational Redundancies

Source Slide Note

The official slide deck's "Integrity Tools" list also includes "Availability," which reads as a slide-formatting carryover from the following topic rather than an actual integrity tool. The table above lists only the tools that genuinely protect integrity.

Vulnerability

A **vulnerability** is a weakness in a security system. Weaknesses can appear in any element of a computer — hardware, operating system, or software — and each asset type (hardware, software, data) has its own characteristic vulnerabilities.

Four Classes of Security Threats

Any threat to a computer system falls into one of four categories, based on how it disrupts the normal flow of information from a source to a destination:

UNAUTHORIZED ACCESS

Interception

An unauthorized party gains access to an asset — e.g. eavesdropping on data in transit, or theft of hardware/software.

ASSET LOST/UNUSABLE

Interruption

An asset of the system becomes lost, unavailable, or unusable — e.g. a Denial of Service attack, or deletion of a file.

ACCESS + TAMPERING

Modification

An unauthorized party not only accesses but tampers with an asset — e.g. injecting malicious code into software.

COUNTERFEIT OBJECTS

Fabrication

An unauthorized party creates counterfeit objects on a computing system — e.g. inserting fake records into a database.

| Physically Securing Your Computer

Software defenses matter little if the hardware itself isn't protected.

- **Computer locks** — cable locks that physically secure a desktop or laptop to a desk, deterring opportunistic theft.
- **Mobile proximity alarms** — a paired transmitter/receiver that triggers an alarm if a device is moved beyond a set range from its owner.
- **Choose your browser deliberately** — most malware arrives through the web browser, so its security track record matters.
- **Download only from trusted sources** — applies to all software, including antivirus programs themselves; a fake "antivirus" download is a classic malware delivery method.

CHAPTER 11

Service Learning: Graphic Design & Canva

CC 101's Service Learning module shifts from computing theory to a **practical, creative skill**: producing professional-looking visual content for real community output — posters, flyers, resumes, certificates, and infographics. It covers the design principles behind good visuals and Canva, the browser-based tool used to produce them.

| What Is Graphic Design?

Graphic design is the art and profession of selecting and arranging visual elements — **typography, images, symbols, and colors** — to convey a message to an audience. Graphic designers plan, analyze, and create visual solutions to communication problems, deciding the most effective way to get a message across in print, electronic, or film media.

What Graphic Designers Produce

Magazines, newspapers, and journals · Corporate reports and publications · Promotional displays and packaging · Commercial displays and billboards · Brochures · Company logos and signs · Webpages and multimedia projects.

The 7 Basic Principles of Graphic Design

Every well-designed layout — a poster, a slide, a resume — applies some combination of these seven principles.

1. Balance

Distributing design elements (shapes, text boxes, images) evenly throughout the layout.

Three types:

Symmetrical (mirrored),
Asymmetrical (uneven but visually weighted),
Radial (arranged around a center point).

2. Alignment

Creates a visual connection between elements — images, shapes, or blocks of text — by lining them up along shared edges rather than scattering them randomly.

3. Hierarchy

Combines dominance and priority, giving extra visual weight to certain elements (bigger headline, bolder color) so the eye knows what to look at first.

4. Contrast

Guides the viewer's attention to key elements by making them visually distinct — e.g. light text on a dark background reads clearly; low-contrast text gets lost.

5. Rhythm

Brings together different elements in a repeated, consistent pattern to create a more organized, cohesive look — irregular, inconsistent spacing feels chaotic.

6. Proximity

Declutters a design by grouping related elements close together, forming a visual connection between items that belong together (vs. scattering them apart).

7. Colour & Space

The right color palette defines the design's tone. Space is the area around or between elements — enough of it (whitespace) keeps a layout from feeling cramped.

| Typography

Typography is the art of arranging letters and text so the copy is legible, clear, and visually appealing. It dates back to the 11th century innovation of movable type, and before the digital age was a specialized craft tied to books, magazines, and public works.

Why Typography Matters

Typography is far more than picking a pretty font — it's a vital part of user interface and visual design. It **builds brand recognition** (a consistent typeface makes a brand instantly identifiable), **influences decision-making** (tone shapes how a message is received), and **holds the reader's attention**.

Choosing the Right Typeface

- **Think about personality** — does the typeface's shape and weight match the message's mood?
- **Reflect on tone** — formal, playful, urgent, elegant?
- **Don't skimp on function** — legibility at the intended size always comes before style.
- **Consider performance** — will it render consistently across devices/platforms?
- **Get inspired** — reference existing well-designed work for direction.
- **Take time to test** — view the typeface at actual size and context before finalizing.

| Introduction to Canva

Canva is a graphic design tool built to simplify digital design. An account is free to create and works via web browser or the iOS/Android apps. Its simple drag-and-drop interface supports image editing and project-based work — from group brainstorming to individual projects — making it well suited for classroom and community-service output.

Getting Started

Sign up at **canva.com** using an email, Google account, or Facebook login. During setup you choose your account type (teacher, student, or other) — this tailors the template and search experience to your needs.

The Canva Workspace

Area	Function
Menu Bar	Top bar — file options, resize, undo, share, and download/publish controls.
Side Panel	Left-hand tool tabs — Templates, Photos, Elements, Text, Music, Videos, Background, Uploads, Folders.
Canvas	The main center workspace where the design itself is built and edited.

Steps for Creating Visual Content With Canva

1 Launch Canva & Sign In

Go to canva.com and sign up or log in (Google sign-in is fastest). Start with a content strategy — identify the goal and platform (e.g. a social post for Facebook/Instagram vs. a poster to print and display locally) before designing.

2 Browse Templates or Set a Custom Size

Canva's template library is organized into categories and subcategories by content type. Alternatively, click **Custom size** in the top-right corner and set width/height in pixels, mm, or inches — useful for blog graphics or non-standard print sizes.

3 Create & Edit the Design

Build the design using the side panel tools (see below). Highlight any text to bring up a formatting bar for size, font, and color adjustments.

Using the Canva Sidebar / Side Panel

Tab	What It Does
Search	Looks for photos, illustrations, or all items; narrow results by clicking Photos or Illustrations.
Elements	Photos, grids, frames, shapes, lines, illustrations, icons, and charts — select a category, then drag and drop onto the canvas.
Grids	

Tab	What It Does
	Placeholder layouts for holding multiple images — pick the grid matching the number and arrangement of images needed.
Text	Pre-formatted headings, subheadings, body text, and ready-made font-pairing combinations.
Background	Solid colors, gradients, patterns, and background images for the whole canvas.
Uploads	Upload and reuse your own images/logos. To remove one, hover over it, click the "i" icon, then the trash icon.

Saving, Downloading & Sharing

Format	Best For
JPG	Smaller file size, general sharing; no transparency support.
PNG (Recommended)	Higher quality, supports transparent backgrounds — Canva's suggested default.
PDF (Standard)	Digital documents meant to be viewed on-screen.
PDF for Print	Print-ready output with correct bleed/quality for physical printing.

Sharing a Design

Click **Share** in the top-right corner: share by **email** (invite specific people to view/edit), **link** (copy a shareable link, with an option to allow editing), or directly to **social media** (Facebook Page, Twitter, Facebook Group) — plus a **Download** option for the formats above. **Embed** is only available for designs set to public.

Logic Gates & Boolean Algebra

This is where binary (Chapter 04) stops being an abstract numbering trick and becomes **physical circuitry**. Logic gates are the actual electronic components — built from transistors — that make decisions and do arithmetic inside every computer. This chapter covers the seven standard gates, their truth tables, Boolean expressions, and the basic electronic components used to build and test them on a breadboard.

Builds On Chapter 04

Logic gates only make sense once binary (0/1, ON/OFF) is second nature. If any of this feels shaky, loop back to **Chapter 04 — Number Systems** first. This chapter, in turn, is the foundation for future subjects like Computer Organization and Digital Logic Design later in the BSIT program.

Digital Logic Circuits & Binary Logic

Digital logic circuits are used to process information inside computers. They operate on binary variables that are symbolically represented by the values **0** and **1** — also described as **low** and **high** logic. **Binary logic** deals with these binary variables and the logic operations applied to them: **ON = Logic 1, OFF = Logic 0**.

The Transistor — The Physical Switch Behind It All

A **transistor** is a special kind of electronic device used for many important functions. In computers, it serves as a switch that is either **ON** or **OFF** — the same ON/OFF idea from Chapter 04's binary discussion, just now realized as an actual physical component instead of a written digit.

Logic Gates & Boolean Algebra

Logic gates are the actual physical components that perform arithmetic operations or make choices in a computer — constructed from a number of primitive electronic elements (transistors). A logic gate takes one or more

binary **inputs (operands)** and produces a single binary **output**, e.g. $A + B = C$ where A and B are operands, "+" is the operator, and C is the output.

Boolean Algebra is an expression or statement whose purpose is to solve logic problems — it is ideally suited to digital circuit analysis. A **Truth Table** is a table of all possible combinations of input variables, showing the relationship between those input values and the resulting output that describes the behavior of a logic gate (Mano, 2001).

The 7 Standard Logic Gates — Reference Table

Gate	Rule	Boolean Expression	IC Chip (74LS Series)
NOT (Inverter)	Output is always the opposite of the input.	$F = A'$	74LS04
BUFFER	Output equals the input exactly — no logic change, used to strengthen/ isolate a signal.	$F = A$	—
AND	Output is HIGH only when all inputs are HIGH.	$F = A \cdot B$ (or AB)	74LS08
OR	Output is HIGH if any or all inputs are HIGH.	$F = A + B$	74LS32
NAND	Output is LOW only when all inputs are HIGH (AND, inverted).	$F = (A \cdot B)'$	74LS00
NOR	Output is HIGH only when all inputs are LOW (OR, inverted).	$F = (A + B)'$	74LS02
XOR	Output is HIGH only when the inputs are different from each other.	$F = A \oplus B$	74LS86

Gate	Rule	Boolean Expression	IC Chip (74LS Series)
XNOR	Output is HIGH only when the inputs are the same — XOR followed by an inverter.	$F = (A \oplus B)'$	74LS26

Watch Out for a Common Slide Typo

Some lecture slides print the NAND expression as $F = A'B'$ — that's actually the expanded form of **NOR** (by De Morgan's Theorem, $(A+B)' = A'B'$). The correct NAND expression is $F = (A \cdot B)' = A' + B'$. If you see $A'B'$ labeled NAND anywhere, it's a mislabel — trust the truth table instead: NAND is only LOW when both inputs are HIGH.

Truth Tables at a Glance

Single-Input Gates

A (Input)	NOT	BUFFER
0	1	0
1	0	1

Two-Input Gates

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Memory Trick

NAND and NOR are simply AND and OR with the output flipped (bubble on the output). XOR = "exclusive OR" — HIGH only when inputs disagree. XNOR = "exclusive NOR" — HIGH only when inputs agree (equality checker).

Worked Examples

Example 1 — Three-Input AND Gate

Consider the expression $A \times B \times C = F$. Implement it using a three-input AND gate (built from two 2-input AND gates in sequence: first $A \cdot B$, then that result ANDed with C) and list all possible input combinations with their output.

A	B	C	AB	F = ABC
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

Only one row out of eight produces a HIGH output — this is the defining trait of AND-type logic: everything has to line up for the output to fire.

Example 2 — $F = X(X + Y)$

Evaluate by determining the binary value of the expression for all possible combinations of the binary input variables.

X	Y	X + Y	F = X(X + Y)
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

Boolean Algebra Insight — The Absorption Law

Notice the **F** column is identical to the **X** column. That's not a coincidence: $X(X + Y) = X$ is one of the standard **absorption laws** of Boolean algebra. Recognizing patterns like this lets you simplify circuits before ever building them.

Breadboards & Basic Electronic Components

A **breadboard** is a circuit board used to make temporary circuits — a device for prototyping electronics and testing circuit designs without soldering anything permanently.

Terminal

BREADBOARD PART

Holds the components

The rows of holes that grip and hold the electronic elements (ICs, wires, resistors) in place.

Bus

BREADBOARD PART

Distributes power

The long rails (usually marked + and –) used to supply electric power to all the electronic components on the board.

Integrated Circuit (IC)

COMPONENT

A chip full of gates

A small black chip that packages one or more logic gates internally. Pin 1 is marked with a notch or dot; pins are numbered counter-clockwise around the chip.

Resistor

COMPONENT

Limits current flow

Color-banded component that restricts the amount of current flowing through a circuit,

Light Emitting Diode (LED)

COMPONENT

Visual output indicator

Lights up when current flows through it in the correct direction —

Transistor

COMPONENT

The switch itself

The underlying electronic switch that logic gates are built from — combined in different

protecting components like LEDs from burning out.

commonly used to visually confirm a gate's HIGH/LOW output on a breadboard circuit.

arrangements inside an IC to create AND, OR, NOT, and every other gate.

Filipino Analogy — Logic Gates as Household Switches

Think of an **AND gate** like a house with two switches wired in series for one light (e.g., a switch at the main door AND a switch at the stairs) — the light only turns on if **both** are flipped ON. An **OR gate** is like two switches wired in parallel for a hallway light — **either** switch alone turns it on. A **NOT gate** is like a "sensor-off" light — flipping the sensor ON actually turns the light OFF, and vice versa.

CHAPTER 13

Web Basics — Internet, the Web & Markup Languages

This is the bridge chapter — the last theory unit before WS 101 turns everything here into hands-on coding. It zooms out from hardware/software (Chapters 03–07) and networking internals (Chapter 08) to look at how the Internet, the Web, browsers, and websites fit together, then introduces the two markup languages — HTML and XML — that WS 101 builds on directly.

| The Internet

The **Internet** is a globally connected network system that uses **TCP/IP** to transmit data across many kinds of media. It's a network of global exchanges — private, public, business, academic, and government networks — linked by guided (wired), wireless, and fiber-optic technologies.

Internet Protocol (IP) is the Internet's primary component and communications backbone. Because the Internet is built from both hardware and software layers, IP is the addressing standard used to identify every

uniquely connected device. The two versions in active use are **IPv4** and **IPv6**.

| The World Wide Web

The **World Wide Web (WWW)** is a network of online content formatted in HTML and accessed via HTTP — all the interlinked HTML pages reachable over the Internet. It's what most people mean when they casually say "the Internet," but the two aren't the same thing: the **Internet is the underlying network connection**; the **Web is the collection of pages, pictures, videos, and content** you reach through a browser using that connection. The early Web was entirely text-based, hosted only by organizations technically capable of running a Web server and writing HTML.

Quick Clarification — Why the Year Changes Depending on the Source

This chapter's official slide says the Web was "originally designed in 1991 by Tim Berners-Lee." Chapter 08 of this guide says "invented in 1989." Both are defensible: Berners-Lee **proposed** the WWW in 1989 at CERN, **built** the first browser and server in 1990, and it went **publicly live** on the wider Internet in 1991. If an exam question cites a specific year, match it to whichever milestone that question is actually asking about.

| The Web as a Communication Medium — 5 Core Components

Networks

INFRASTRUCTURE

The physical backbone

Local-area and wide-area networks connecting computers worldwide — together, these form the Internet.

Clients

SOFTWARE

The user's window to the Web

Web browsers that let end-users actually access and view Web content.

Servers

SOFTWARE

Always-on providers

Constantly running programs that serve information and pages to the Web on request.

Documents

CONTENT

What actually gets delivered

Protocols

RULES

How everyone agrees to communicate

Web pages, mostly coded in HTML, that supply the information visitors came for.

HTTP is how Web clients and servers talk to one another; it depends on the underlying TCP/IP protocol suite.

What the Internet Offers

As the most cost-effective communications method in the world, the Internet makes these services instantly available:

- Email
- Web-enabled audio/video conferencing
- Online movies and gaming
- Data transfer / file-sharing, often via File Transfer Protocol (FTP)
- Instant messaging
- Internet forums
- Social networking
- Online shopping
- Financial services

Ways to Connect to the Internet

Every personal device connects using different hardware, each with its own realistic speed range. The common options:

Connection Type	How It Works
Wireless (Wi-Fi)	Connects to a local router over radio signal — the standard for homes, schools, and offices.
Mobile (cellular data)	Uses a phone's SIM/cellular network (3G/4G/5G) — no fixed router needed.
Hotspot	A phone or portable device shares its own mobile data connection wirelessly with other devices.
Dial-up	Uses a phone line and modem to connect — the earliest home method, now essentially obsolete.

Connection Type	How It Works
Broadband	General term for any always-on, high-capacity connection — cable, fiber, and DSL all fall under this umbrella.
DSL	Digital Subscriber Line — delivers broadband over existing copper telephone lines.
Satellite	Connects via signal to an orbiting satellite — used where cable/fiber infrastructure isn't available (e.g. remote provinces).

Web Browsers

1990

WorldWideWeb (later renamed Nexus)

Created by Sir Tim Berners-Lee — the first web browser, giving people a basic way to view web pages.

1992

Mosaic

The first popular **graphical** browser. Before Mosaic, the Web was text-based, plain, and needed technical know-how to use at all.

Modern browsers add features like **tabbed browsing** — many pages open in individual tabs instead of separate, resource-heavy windows. The five browsers this course lists as most popular:

- Mozilla Firefox
- Google Chrome
- Apple Safari
- Opera
- Internet Explorer

Outdated by 2026 — Know It for the Exam, Not for Real Use

Microsoft officially retired Internet Explorer in June 2022, folding its remaining functionality into **Microsoft Edge**. It's still fair game as a recall answer since it's what the official slide lists, but don't actually go looking for it on a modern device.

What Is a URL?

A **Uniform Resource Locator (URL)** is the address of a resource on the Internet — it indicates both the resource's location and the protocol used to access it.

EXAMPLE URL

`https://www.example.com`

Part	Segment	Role
Protocol (scheme)	https://	The access method used to reach the resource
Sub-domain	www.	Optional prefix pointing to a specific service on the domain
Domain name	example	The registered, human-readable name of the site
Top-level domain (TLD)	.com	The category/region suffix — .com, .edu, .ph, .gov, etc.

Components of a Website — Page Layout Top to Bottom

Regardless of purpose or design, most websites share a common architecture. Reading a page top to bottom, that architecture usually looks like this:

1 Header/Banner TOP OF PAGE

Holds the logo of the company, publisher, or owner — instantly tells visitors what the site is about.

2 Navigation Bar MENU / TABS

Lets visitors reach other pages; appears on every page for consistent navigation.

3 Sidebar OPTIONAL

Increasingly dropped by modern sites — often seen as a distraction, especially when it holds ads or social feeds.

4 Content MAIN BODY

The core information/ media the site exists to deliver; content quality usually decides the site's actual value.

5 Footer BOTTOM OF PAGE

Reached only by visitors who stayed engaged; typically holds links, copyright, and disclaimers.

Front-End vs. Back-End Elements

WHAT VISITORS SEE

Front-End Elements

Page layout, logo, images/graphics, the navigation structure (the order of pages — what links to what), and graphic design as a whole. Everything a visitor directly sees and interacts with.

WHAT RUNS UNDERNEATH

Back-End Elements

Some sites are fully static — the same every visit, with no polls, bulletin boards, or referral forms. A **Content Management System (CMS)** lets an owner update content without directly editing the HTML.

Other Website Features

Feature	What It Does
E-Commerce	Lets visitors purchase items directly — extends a small merchant's reach to a worldwide audience.
Shopping Cart	For a catalog of just one or two products, a simple click-fill-purchase flow is enough.
Site Search	Lets visitors find specific content with a quick search instead of browsing manually.
Blog Feature	Regularly updated articles, run independently or as part of the main site.
Image-Rotation	Presenting new images on repeat visits gives the page a sense of life.
Multi-Media	Audio, video, and interactive content beyond plain text and static images.
Security	Protects trade secrets, proprietary code, and sensitive visitor data such as credit card numbers.

Languages of the Web

A Web browser talks to a Web server through **Hypertext Transfer Protocol (HTTP)**, built to handle hypertext and hypermedia documents — regular text, images, audio, and video. Native Web pages are written in **Hypertext Markup Language (HTML)** and saved with the `.html` (or `.htm`) suffix.

Language	Full Name	Role
HTML	HyperText Markup Language	Defines the meaning & structure of web content — the most basic building block of the Web
XML	Extensible Markup Language	Defines rules for encoding documents in a format that's both human- and machine-readable
JavaScript	—	Describes a page's functionality/behavior
CSS	Cascading Style Sheets	Describes a page's appearance/presentation
PHP	PHP: Hypertext Preprocessor	Server-side scripting language for dynamic, database-driven pages

Forward Link — This Is WS 101's Entire Roadmap

WS 101 builds these five languages in this exact order: **HTML** (Weeks 2–3) → **CSS** (Week 4) → **JavaScript** (Weeks 5–6) → **PHP + MySQL** (Weeks 10–11) → **Laravel** (Weeks 12–14). Everything in this table is vocabulary you'll be typing, not just memorizing, within a few weeks.

HTML — HyperText Markup Language

HTML is the most basic building block of the Web. It defines the **meaning and structure** of web content. Other technologies are generally used to

describe a page's **appearance/presentation** (CSS) or its **functionality/behavior** (JavaScript) — HTML itself only handles structure.

| XML — Extensible Markup Language

XML is a markup language that defines a set of rules for encoding documents in a format that's both human-readable and machine-readable. Its design goals emphasize simplicity, generality, and usability across the Internet — it's a textual data format with strong Unicode support for different human languages.

| HTML vs. XML — Key Differences

FIXED VOCABULARY

HTML

Uses a fixed set of **predefined tags** (<p>, <div>, <table>...) — you can't invent new elements. HTML itself **is** a markup language.

CUSTOM VOCABULARY

XML

Provides **provisions for defining new elements** — you create your own custom tags to describe data. XML can be used to **build** other markup languages.

| Web Hosting

Web Hosting is a service that stores and serves ready-made files and programs so they're accessible on the Web. Publishing a website on the Web means:

1 Design & construct the site

Build the pages and write the underlying programs for the website.

2 Place it with a hosting service

Colleges and universities often host personal/educational sites for students and faculty free of charge; commercial web hosting companies provide the service for a fee.

Characteristics of a Good Website

Source Material Note

The official slide deck titles this section "10 Characteristics," but only **8** are actually listed in the material below. Treat this list as incomplete — flag it with your instructor if an exam expects all 10.

Characteristic	What It Means
Functional	Serves a specific purpose, usually solving a problem — e.g. a job board lets employers post jobs and seekers apply.
Easiness	Should be easy to use and navigate.
Relevant Content	Content should stay relevant to users — no car listings on a job board.
Modern	Should use current design trends and technologies, not support outdated needs.
Performance and Speed	Should load fast enough to show visitors what they're looking for.
Reliable	Should reliably deliver what it promises — e.g. on-time notifications after a job application.
Scalable	Should have infrastructure sufficient to support a growing number of visitors.
Secure	Should follow industry security standards and guidelines to protect visitor data.

Filipino Analogy — A Website as a Sari-Sari Store

The **Header/Banner** is the signage above the door — it tells you whose tindahan this is before you even step in. The **Navigation Bar** is the aisles or sections. The **Content** is the paninda itself — the goods customers actually came for. The **Footer** is the resibo at the counter — the last thing you interact with, holding the fine print like store hours and contact info.

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